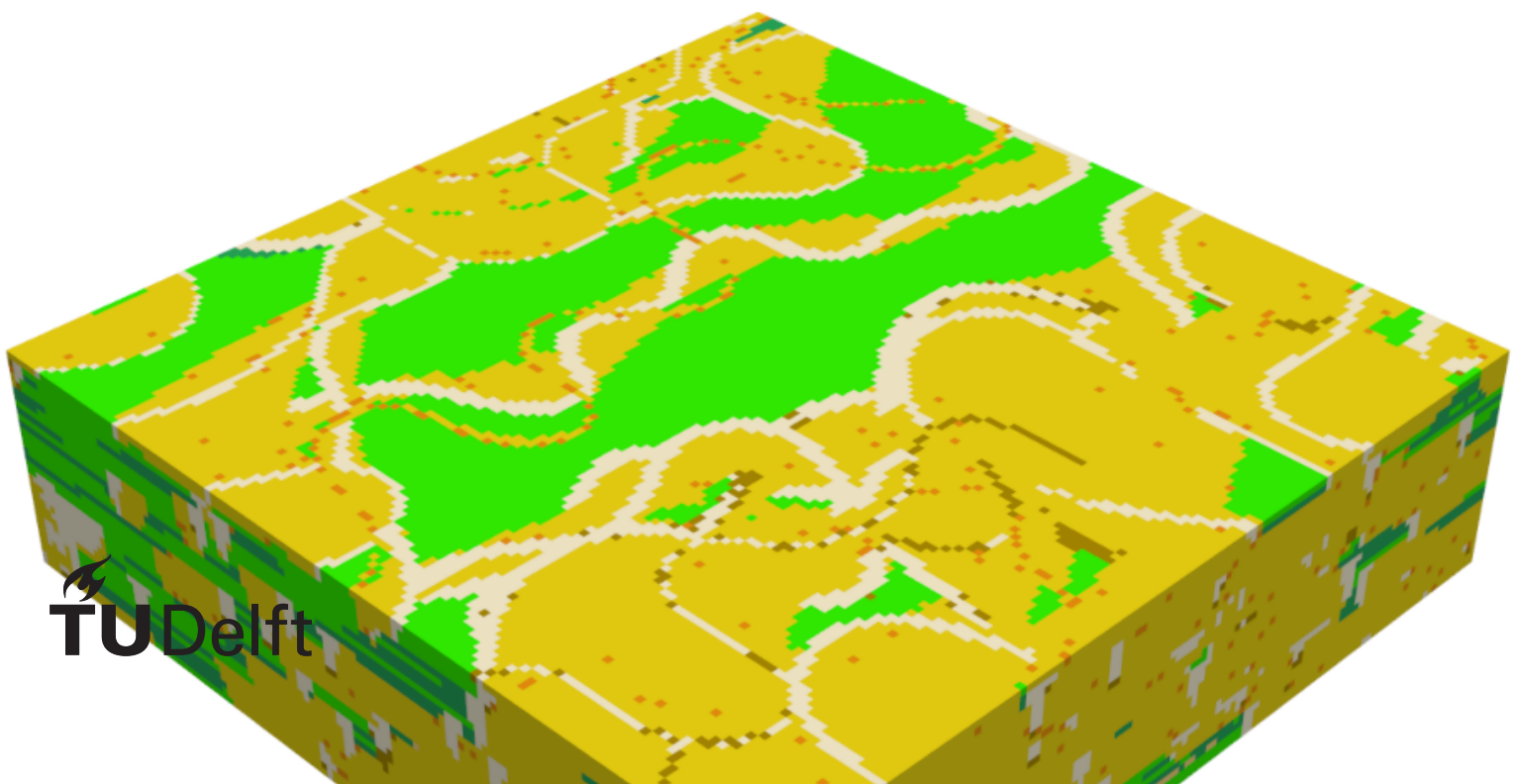


Evaluating the applicability of Generative Adversarial networks to generate geological facies models

A Case study for the Delft Area

M.D. Torres Ruhe



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by

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An electronic version of this thesis is available at <http://repository.tudelft.nl/>.

Abstract

Here is the text of my abstract...

Acknowledgements

I would like to thank...

List of Abbreviations

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cGAN Conditional Generative Adversarial Network. 11

DSST Delft Sandstone. 3, 4

FluvGan Fluvial Generative Adversarial Network. 11

GAN Generative Adversarial Network. 11

TUD Technische Universiteit Delft. 3

TvD True Vertical Depth. 3

WNB West Netherlands Basin. 3

Contents

Abstract	i
Acknowledgements	ii
1 Introduction	1
1.1 Problem definition	1
1.2 Research gap and opportunity	2
1.3 Objective and research questions	3
1.4 Thesis outline	3
2 Background	4
2.1 The Delft Geothermal Reservoir	4
2.1.1 Geology of the Delft Reservoir	4
2.1.2 Deltaic Depositional Environment.	5
2.2 Geological Modelling	5
2.3 Machine Learning	6
2.3.1 Deep Learning and Inductive Biases.	8
2.3.2 Geomodelling with Deep Learning	9
2.3.3 Generative Adversarial Networks	9
3 Methodology	11
3.1 Dataset Generation	11
3.1.1 Dataset Descriptions	11
3.1.2 Sample preprocessing	11
3.2 Fluvgan Framework	12
3.2.1 Model Architecture	12
3.2.2 Optimization & Inversion	14
3.3 Evaluation Metrics	14
4 Results	16
4.1 Datasets	16
4.2 Post Training Generative Results	17
4.2.1 Hyperparameter Tuning	17
4.2.2 cGAN Unconstrained Performance	19
4.2.3 High-Fidelity 9-Class Generative Performance	21
4.3 Post-Inversion Results	22
5 Discussion & Recommendations	24
5.1 Interpretations of key results	24
5.2 Limitations of the current work	24
5.3 Recommendations for future work	24
6 Conclusion	26
6.1 Answers to Research Questions	26
6.2 Final Remarks	26
A FLUMY dataset generation	30
B FluvGAN	32
B.1 One Hot Encoding Strategy	32
B.2 Fluvgan Architecture	32

C	Mathematical Formulations for Optimization and Evaluation	34
C.1	Optimization & Inversion Framework.	34
C.1.1	Latent Space Optimization.	34
C.1.2	Pivotal Tuning	35
C.2	Generative Evaluation Metrics.	35
C.2.1	Wasserstein Distance and Mean Sliced Wasserstein Distance	35
C.2.2	Uncertainty and Spatial Divergence Metrics	36

1 Introduction

Sedimentary deposits represent a vital commodity in modern society due to the various resources that can be extracted from them. These resources include minerals, groundwater, geothermal heat, sands and hydrocarbons (e.g., Brekke et al., 2001; Donselaar et al., 2017; Donselaar et al., 2015; Morris & Ramanaidou, 2007). Moreover, recent studies have investigated storage possibilities for carbon dioxide and hydrogen in this subsurface environment (Heinemann et al., 2021; Zhou et al., 2020). Beyond the increase in subsurface applications, the world also deals with an increasing population and energy demand. Consequently, the demand for resources hosted within fluvial sedimentary deposits is expected to increase while the distribution of the commodities themselves becomes sparser and of lower quality (e.g., Calvo et al., 2016; Spittler et al., 2020). This highlights the need for a deeper understanding of subsurface environments to enable more precise resource management which can be achieved through various methods such as: data gathering surveys, geological evolution and constructing high-quality 3D geomodels.

3D geomodels are traditionally constructed by engineers and geoscientists to predict reservoir properties and resource distribution (Royer et al., 2015; SONG et al., 2022). The voxels in these models may be populated by continuous or categorical properties: grain size, porosity & permeability values or facies associations. These predictions are fundamentally based on a foundational geological conceptual model. The conceptual model acts as a hypothesis regarding the most likely rock mass configuration, built entirely from sparse field data (Koltermann & Gorelick, 1996). This data sparsity is an unavoidable reality of subsurface exploration, resulting from the exceptionally high costs and extended runtimes required for physical data acquisition. Consequently, engineers develop a working understanding of the subsurface based on limited information. This reliance on limited information drives the continuous need to improve geomodelling techniques. Such improvements aim to computationally bridge the gap between isolated data points and realistic, continuous geological representations.

1.1. Problem definition

Despite ongoing advancements in data collection, data scarcity dictates that functional 3D geomodels must rely heavily on subjective interpretations of the underlying geological conceptual model. Because transitioning from these interpretive hypotheses into a digital framework requires applying specific mathematical and spatial algorithms, this workflow inevitably introduces uncertainty at multiple stages, stemming from both the sparse data inputs and the modelling steps themselves (Høyer et al., 2024). As a result, precisely measuring its extent is crucial for effective decision-making and can be accomplished with various techniques (e.g. Monte Carlo Simulations), necessitating the creation of a large collection of equally likely 3D geomodels. Creating this extensive ensemble, however, reveals a core drawback in conventional modeling techniques: the significant compromise between computational efficiency and geological precision. To sustain the generation rate needed for extensive uncertainty quantification, traditional methods frequently depend on greatly simplified, weak geological assumptions.

Due to the fact that these approaches are often limited to simple statistical distributions or strict object-oriented rules (Rongier & Peeters, 2025b), they find it challenging to replicate the intricate, non-linear architectural relationships present in natural sedimentary formations (Sun et al., 2023). As a result, existing workflows compel engineers to decide between models that can quickly quantify uncertainty but do not possess geological realism, or highly precise multi-conceptual models that require too much effort to produce in large quantities (Enemark et al., 2019). This bottleneck highlights a significant deficiency in contemporary subsurface characterization. A significant need exists for a modeling approach that can thoroughly incorporate complex geological prior knowledge while also ensuring the fast generation rates necessary for ensemble-based uncertainty quantification. As a result, the inquiry comes up regarding how the existing modelling process can be modified to ensure that:

- 1) Previous geological insights and structural intricacies are better employed.

- 2) Uncertainty in the produced models is quantified more precisely and efficiently.

To address the need for both geological realism and rapid ensemble generation, recent research has proposed advanced machine learning algorithms as a highly effective solution. These algorithms, specifically the modern subset known as Deep Learning, possess the unique ability to learn complex, non-linear spatial relationships directly from data. By learning these intricate relationships, Deep Learning can accurately capture the geological complexity and non-stationarity that traditional methods often miss (Rongier & Peeters, 2025b). Furthermore, once such a model is fully trained and validated, the generative subset of Deep Learning models can generate a vast ensemble of 3D realizations at a fraction of the time compared to slower process-based models. This rapid generation directly facilitates the large-scale uncertainty quantification demanded by modern subsurface engineering.

While the broader application of machine learning in geomodelling has historical roots (Demyanov et al., 2008), recent surges in computing power have yielded highly capable 3D **Deep Generative Modelling (DGM)** architectures, including **Variational Auto Encoders (VAEs)** and **Latent Diffusion Models** (e.g., Bhavsar et al., 2024; Di Federico & Durlofsky, 2025; Laloy et al., 2017; Song et al., 2023). Among these varied architectures, this study specifically focuses on **Generative Adversarial Networks**. While recent literature highlights **Latent Diffusion Models** as a vanguard for precise hard data conditioning, they inherently suffer from slow inference speeds due to their iterative denoising process. In contrast, **Generative Adversarial Networks** offer near-instantaneous sampling speeds perfectly suited for large-scale Monte Carlo uncertainty loops, despite well-known training vulnerabilities such as mode collapse. **Generative Adversarial Networks** are therefore selected due to their computational efficiency, established research history in spatial generation, and the wide availability of architectures strictly tailored for geomodelling. However, successfully training such **Generative Adversarial Networks (GANs)** requires an extensive volume of high-quality data to act as the geological prior. To overcome the inherent scarcity of real subsurface data, process-based forward modelling software, such as **FLUMY** (Martin et al., 2016), can be utilized to generate robust synthetic training datasets.

1.2. Research gap and opportunity

Utilizing these synthetic datasets allows **Generative Adversarial Networks** to effectively learn geological priors; however, a critical research gap remains regarding their ultimate deployment. While **DGM** excels in theoretical environments, recent literature emphasizes an urgent need to transition from idealized, synthetic test cases to practical field applications. As Rongier and Peeters (2025b) stated, “We now need to move towards more detailed studies on larger case studies closer to field applications to further assess how the whole workflow performs”. Similarly, Di Federico and Durlofsky (2025) noted that the application (and extension as required) of the overall methodology to real cases should be performed. Addressing this explicit mandate forms the core opportunity of this research. Consequently, the primary goal of this project is to bridge the divide between synthetic training environments and practical field operations by assessing the performance of deep generative modelling in a more complex, real-world application.

Add: section with the full opportunity for this study

Testing these models in a real-world application requires targeting an active subsurface system, a role fulfilled in this study by utilizing data from the Delft Sandstone reservoir. This reservoir target must be characterized using a simple geological conceptual model combined with sparse well data. Such sparse well data will serve as the hard conditioning constraints to evaluate how the generative model behaves under tight physical limitations. These limitations, and the resulting fluctuations in uncertainty, will be analyzed by testing the workflow across multiple distinct geological priors. Investigating the interplay between varied priors and sparse conditioning constraints ultimately establishes the framework for the specific objectives of this study.

1.3. Objective and research questions

With the research gaps in mind, the main goal of this thesis will be to assess the applicability of using a Generative Adversarial Network to generate realizations of a real-world reservoir by conditioning it to well data. Based on the outcome of this thesis, new insight can be given into the workflow needed to approach applying and evaluating Generative Adversarial Networks for generating subsurface reservoir realizations. Hence, the main research question of this work is: *To what extent can Generative Adversarial Networks be used to create geologically plausible realizations of the Delft Sandstone reservoir while conditioned to sparse real-world data?* To help answer this question, several sub-research questions have been formulated:

1. How should the Delft Sandstone's geology be parameterized in FLUMY to produce a training dataset sufficiently diverse for DGM modeling?
2. Which combination of hyperparameters yields a stable training process?
3. To what degree is the model able to reproduce the complexity inherent in the training data?
4. To what degree does the model generate realizations that are successfully conditioned to well data?
5. How different do models trained on different geological priors react when sparse data is introduced?

1.4. Thesis outline

The remainder of this thesis is structured to systematically address these research objectives. Chapter 2 establishes the theoretical foundation by reviewing the principles of geological modelling, the mechanics of deep generative models, and the specific geological setting of the Delft Sandstone Member. Building upon this theoretical background, Chapter 3 outlines the core workflow, describing the generation of synthetic training datasets using FLUMY and detailing the specific Generative Adversarial Network (GAN) architecture deployed for the reservoir modelling tasks. The empirical outcomes of this workflow are presented in Chapter 4, which evaluates the training process and analyzes the structural and spatial realism of the generated realizations. These results are critically examined in Chapter 5, which contextualizes the findings, addresses the inherent limitations of the proposed approach, and highlights avenues for future research. Finally, Chapter 6 summarizes the main contributions of this thesis and provides definitive answers to the central research questions.

2 Background

This chapter establishes the theoretical and geological foundations and reasoning upon which this research is built. It provides a short overview of the geological setting of the Delft reservoir and a literature review of modern geomodelling and machine learning (ML) techniques. Section 2.1 introduces the geological context of the Delft reservoir, detailing its depositional environment and current conceptual model. Section 2.2 explores the state-of-the-art in geological modelling, comparing traditional techniques and their respective limitations. Finally, Section 2.3 provides a general overview of machine learning and deep learning (DL) principles, focusing on their historical evolution and current application to subsurface characterization. This balanced overview is intended to provide both geological and computer science readers with the necessary context to understand the research methodology and results presented in subsequent chapters.

2.1. The Delft Geothermal Reservoir

The data utilized in this study is sourced from the Delft geothermal reservoir, situated beneath the [Technische Universiteit Delft \(TUD\)](#) Campus in the Netherlands. As of February 2026, a geothermal doublet is actively producing heat from this reservoir (Beerda, 2026). A doublet system consists of a two-well setup: a producer well extracts warm formation water, and an injector well returns the cooled water to the reservoir. At the Delft site, the producer well (**DEL-GT-01**) was drilled to a [True Vertical Depth \(TvD\)](#) of approximately 2,337 m, while the injector well (**DEL-GT-02-S2**) reached a TvD of approximately 2,100 m (Vardon et al., 2024). The project serves both commercial and research objectives, acquiring extensive geophysical logs and core data, supplemented by surface and downhole monitoring instrumentation (Vardon et al., 2020; Vardon et al., 2024). Despite the high data density, the cored section is restricted to the first 80 m of the **DEL-GT-01** well (approximately one-fourth of the reservoir thickness), while **DEL-GT-02-S2** contains no cored sections. The primary target is the [Delft Sandstone \(DSST\)](#), a member of the Lower Cretaceous Nieuwerkerk Formation in the [West Netherlands Basin \(WNB\)](#) (Donselaar et al., 2015).

2.1.1. Geology of the Delft Reservoir

The [WNB](#) is a 60 km wide transtensional basin in the southwest of the Netherlands, characterized by NW-SE trending structural highs and depressions (Donselaar et al., 2015; Ziegler, 1990). The [WNB](#) entered an active rifting stage during the Middle Jurassic (Bodenhausen & Ott, 1981; Den Hartog Jager, 1996; Willems et al., 2020), leading to the formation of several half-grabens filled with terrestrial sediments sourced from neighboring regions (Den Hartog Jager, 1996). The progressive filling of these half-grabens coincided with a change in the depositional environment as a transgressive shoreline moved the system from fluvial to marine. The Nieuwerkerk Formation comprises three members: the Alblasserdam, the [DSST](#), and the Rodenrijs Claystone.

The Alblasserdam member was deposited in a fluvial environment characterized by stacked channels and floodplain deposits (DeVault & Jeremiah, 2002). The [DSST](#) directly overlays the Alblasserdam and consists of thick, stacked fluvial-deltaic channel sandstones with high net-to-gross values, oriented NW-SE (Bruining, 2023; Groenenberg et al., 2010; Mijnlieff, 2020; Willems et al., 2020). The [DSST](#) is in turn overlain by the Rodenrijs Claystone, an organic-rich lagoonal deposit (Bruining, 2023; DeVault & Jeremiah, 2002; Willems et al., 2020).

Based on the sedimentological analysis of the **DEL-GT-01** core and **DEL-GT-02-S2** sidewall samples, three primary facies associations (FAs) have been identified (Bruining, 2023):

- **FA1: Fluvial and distributary channel deposits.** Primarily light grey, fine- to medium-grained sandstones featuring low-angle cross-bedding and ripple lamination, with unit thicknesses ranging from 5 to 80 m.

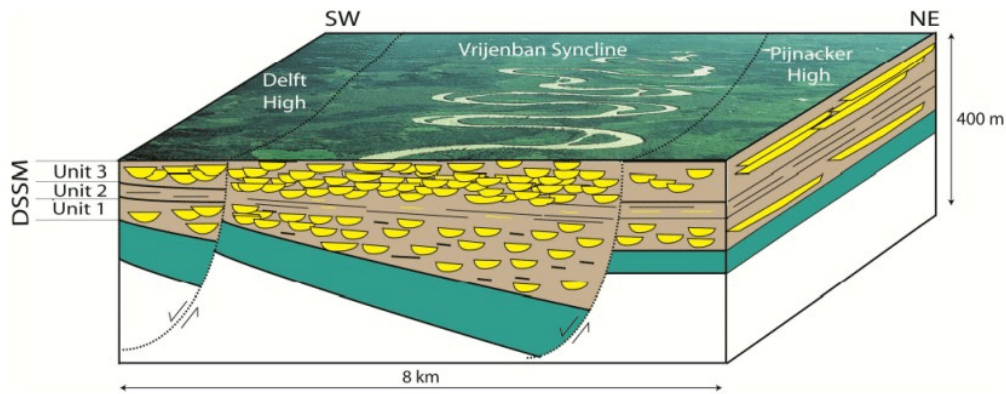


Figure 2.1: Depositional models for the DSST Member with three identified units. Accommodation space is increasing towards the NE (Donselaar et al., 2015)

- **FA2: Crevasse splays and levee deposits.** Composed of moderately sorted, very fine- to medium-grained sandstones interbedded with dark grey siltstones, with thicknesses ranging from 1 to 5 m.
- **FA3: Overbank deposits and paleosols.** Consists of dark grey siltstones, claystones, and very fine-grained sand deposited in low-energy environments. This association is characterized by abundant organic material, slickensides, and siderite nodules.

The specific depositional environment of the DSST remains a subject of scientific discussion. It has been attributed to both fluvial-meandering (Donselaar et al., 2015; Vondrak et al., 2018; Willems et al., 2020) and fluvio-deltaic systems (Bruining, 2023; van Adrichem Boogaert & Kouwe, 1993; Weert et al., 2024). Recent analysis by Bruining (2023) suggests that the cored section represents a deltaic environment. The observed facies associations can be attributed to a upper to lower plain delta with multi-stacked fluvial/distributary channels, single channels, crevasse splays, levees and overbank fines.

2.1.2. Deltaic Depositional Environment

Deltaic environments form at the land-sea interface where river systems supply clastic sediments. Delta morphology is governed by the relative influence of fluvial, wave, and tidal processes (Kolb & Van Lopik, 1966). The subaerial delta plain is typically divided into two domains (Heldreich et al., 2017; Kolb & Van Lopik, 1966):

- **Upper Delta Plain:** Lies above the limit of saltwater intrusion. Sedimentation is driven by distributary channel migration, overbank flooding, and crevasse splaying. Environments include meandering and braided channels, lacustrine fills, and floodplains.
- **Lower Delta Plain:** Subject to river-marine interaction and tidal influence. Channels are more numerous and exhibit bifurcating or anastomosing patterns. Interdistributary bays, bay fills (crevasse splays), and marshes dominate the landscape.

In the DSST, FA1 features organic material and siltstones indicating slackwater periods, while the lack of bioturbation suggests high sedimentation rates. The fine-grained nature of FA2 and FA3 reflects more distal positions on the delta plain. Siderite nodules in FA3 indicate anoxic, waterlogged conditions (Samuel, 1959), while soft-sediment deformation suggests rapid loading of mudstones over saturated sands. These features collectively indicate that FA3 represents deposition within a delta plain, likely in an interdistributary bay or a pro-delta environment.

2.2. Geological Modelling

The primary objective of 3D reservoir modelling is to provide a representative volume of the subsurface for flow simulation and decision support. These models are dynamic, requiring updates as new data becomes available to refine predictions over the reservoir's lifetime (Emerick, 2025; Ringrose & Bentley, 2016). Existing techniques are generally categorized into three types (Koltermann & Gorelick, 1996):

- **Pixel-based Models:** Use spatial statistics (e.g., variograms) for cell-by-cell property assign-

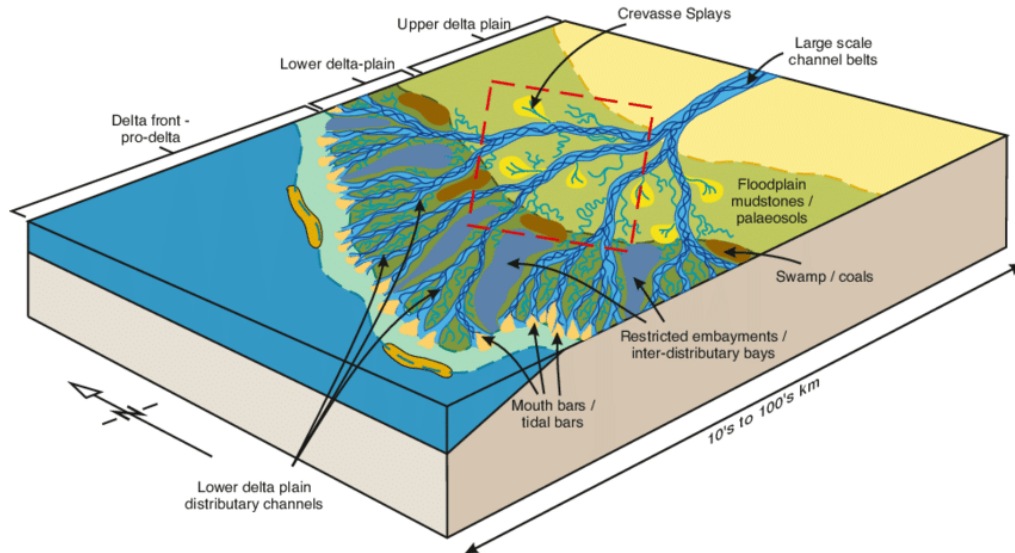


Figure 2.2: Depositional model of the Mungaroo Formation, illustrating the geometries and domains of a fluvial-deltaic system (Heldreich et al., 2017).

ment. They are efficient and honor dense well data but often fail to capture complex connectivity, resulting in pixelated realizations.

- **Object-based Models:** Directly place geometric features (e.g., channels) based on geological rules. They offer high visual realism but struggle to honor dense or closely spaced well data.
- **Process-based Models:** Simulate the processes behind sediment transport and deposition. They provide the highest degree of realism but are computationally expensive and difficult to condition to specific hard data like well logs.

Attribute	Geostatistical	Object-Based	Process-Based
Geological Prior	Variograms, training images.	Geometric shapes and rules.	Physical boundary conditions.
Realism	Low to Moderate.	Moderate to High.	Very High.
Key Advantages	Computational speed, well conditioning.	Realistic connectivity.	Physical consistency.
Key Disadvantages	Lack of geological realistic shapes.	Difficult conditioning.	High computational cost.

Table 2.1: Comparison of spatial modelling techniques in geomodelling.

While traditional pixel-based geomodelling relies heavily on two-point spatial statistics, such as variograms, to distribute properties, these methods often fail to capture the complex, multipoint connectivity inherent in natural geological systems. Figure Machine learning introduces a paradigm shift by moving away from user-defined spatial rules. Instead of requiring an engineer to pre-define the shape of a channel or the range of a variogram, ML algorithms learn these intricate, non-linear spatial relationships directly from data. This makes ML exceptionally well-suited for geomodelling, where the relationships between sparse hard data (e.g., well logs) and the broader geological architecture are too complex to be captured by standard statistical interpolation.

2.3. Machine Learning

Machine learning (ML) involves the development of algorithms that learn patterns from data to perform specific tasks without explicit programming (Goodfellow et al., 2016). This foundational concept

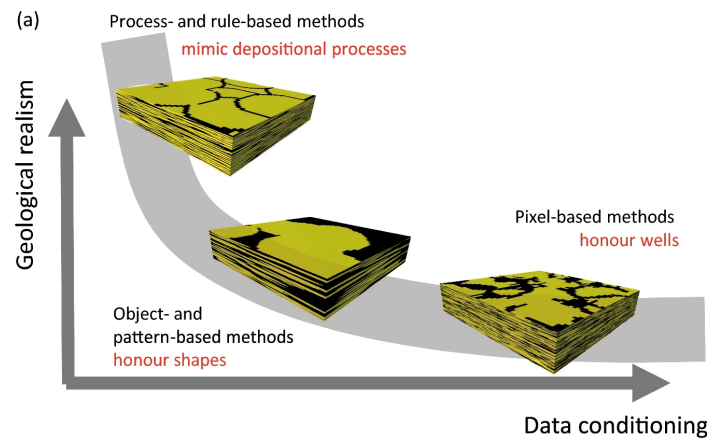


Figure 2.3: Simple overview of different geological modelling techniques compared to their ability to generate geological plausible data and honour well data (Manzocchi & Walsh, 2026). Where process- and rule- based methods are great at generating geological plausible samples but struggle with honouring to well data, pixel-based methods are opposite in nature. Object based models offer a best of both worlds solution.

of learning without explicit programming has evolved significantly since Samuel (1959) introduced the term, expanding into a wide array of computational techniques. These diverse techniques are fundamentally categorized based on their training approach into two primary domains:

- **Supervised Learning:** The algorithm is trained on a labeled dataset containing explicit input-output pairs.
- **Unsupervised Learning:** The algorithm identifies hidden patterns within unlabeled data without prior knowledge of the desired output.

Within these two broad categories, algorithms are deployed to solve common tasks such as classification, regression, clustering, and generation. Applying these generative tasks to the subsurface highlights a severe limitation in traditional pixel-based geomodelling, which relies heavily on simple two-point spatial statistics (e.g., variograms) to distribute properties. Because these simple statistics often fail to capture the complex, multipoint connectivity inherent in natural geological systems, ML introduces a necessary paradigm shift. This shift moves the modelling workflow away from strict, user-defined spatial rules. Instead of requiring an engineer to pre-define the range of a variogram, ML algorithms learn intricate, non-linear spatial relationships directly from the data. Learning directly from data makes ML exceptionally well-suited for geomodelling, where the relationships between sparse hard data (e.g., well logs) and the broader geological architecture are simply too complex to be captured by standard statistical interpolation. The capacity to uncover such complex, non-linear relationships in vast datasets is largely driven by modern [Artificial Neural Networks \(ANNs\)](#) (LeCun et al., 2015). Inspired by the biological structure of the brain, these [ANNs](#) consist of interconnected artificial neurons that process information across multiple computational layers (Goodfellow et al., 2016).

While [ANNs](#) and traditional machine learning algorithms excel at analyzing tabular petrophysical data for various subsurface tasks, they inherently lack actual geological reasoning skills. Without such reasoning to naturally make sense of areas with low data quality, these algorithms are restricted to inferring abstract features directly from their training data (Schaaf & Bond, 2019). Inferring these features in standard ML workflows necessitates manual feature extraction, a process where engineers must explicitly define the relevant geological parameters prior to training. This strict reliance on manual feature extraction creates a critical limitation when attempting to scale these algorithms to 3D geomodels. Within these high-resolution 3D geomodels, which typically contain millions of voxels, manually defining multi-scale spatial features ultimately becomes computationally impractical.

Deep Learning (DL) overcomes this limitation through automatic feature extraction. By utilizing multi-layered networks, DL architectures natively process high-dimensional spatial grids, bypassing the manual extraction bottleneck. For subsurface characterization, this means a DL model can ingest raw 3D conceptual models or process-based simulations and autonomously identify the underlying geological

rules. This capability to scale with high-dimensional data makes DL the premier mathematical framework for modern reservoir modelling.

2.3.1. Deep Learning and Inductive Biases

Deep learning (DL) is a subset of ML focusing on multi-layered neural networks that represent data with multiple levels of abstraction (Goodfellow et al., 2016; LeCun et al., 2015). Achieving these levels of abstraction in high-dimensional spaces introduces a fundamental challenge known as the “curse of dimensionality.” This curse of dimensionality describes the exponential increase in data required as the spatial dimensions grow, making high-dimensional estimation computationally challenging. To overcome this computational challenge and help models generalize from limited data, engineers implement specific architectural choices known as inductive biases (Prince, 2023). A premier example of such an inductive bias is the foundational architecture used for processing spatially structured data: the **Convolutional Neural Network (CNN)**.

CNNs leverage spatial inductive bias to become highly effective for 2D and 3D geomodelling tasks. Executing these spatial tasks relies on a core architectural innovation known as the convolutional layer. This convolutional layer replaces the fully connected neurons of a standard **ANN** with a localized set of learnable filters, commonly referred to as kernels. These multi-dimensional kernels operate by systematically sliding across the input data in discrete increments dictated by a predefined step size, or stride. This sliding stride is frequently combined with padding, a technique that adds artificial zero-value borders to the input to ensure the entire spatial region is comprehensively analyzed. Analyzing this spatial region involves computing a local dot product between the kernel weights and the input values, a mathematical operation visually illustrated in Figure 2.4.

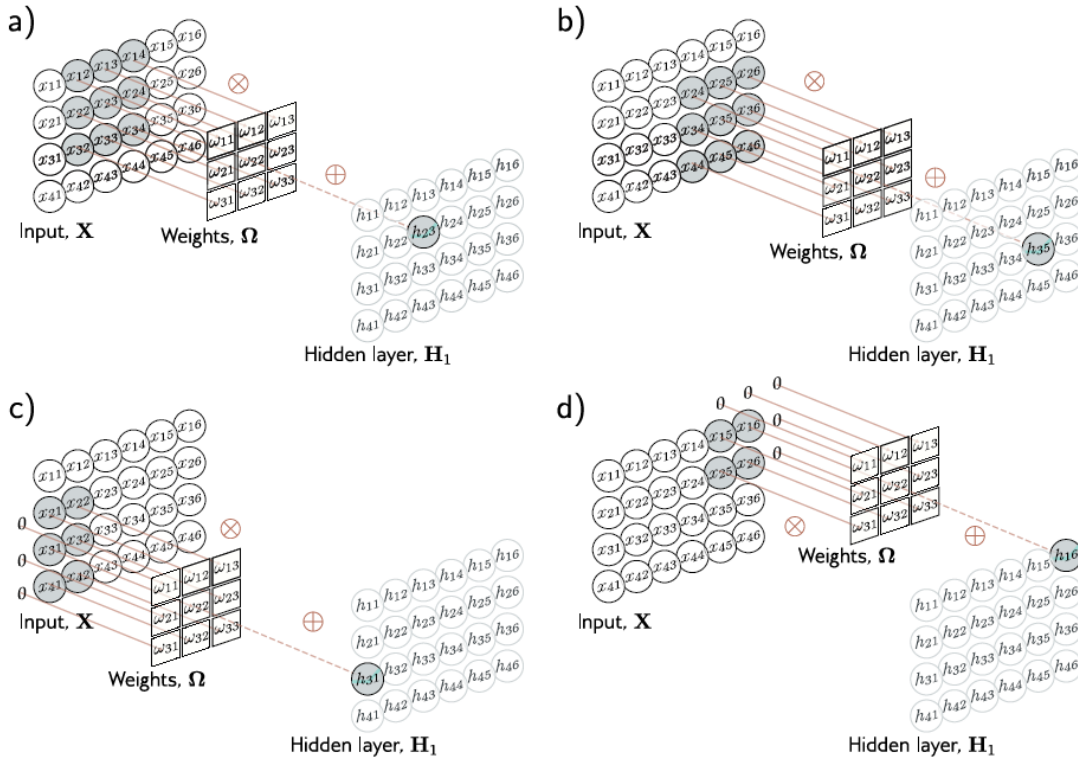


Figure 2.4: Example of a 2D convolutional layer with kernel size of 3x3 and a padding of 1. Each output h_{ij} is composed of a weighted sum of the 3x3 nearest inputs on the previous layer (Prince, 2023). a) Here, the output h_{23} (shaded output) is a weighted sum of the nine positions from x_{12} to x_{34} (shaded inputs). b) Different outputs are computed by translating the kernel across the image grid in two dimensions. c–d) With zero-padding, positions beyond the image’s edge are considered to be zero.

The mathematical result of this operation is a new, filtered spatial representation known as a feature

map. This feature map is typically downsampled by a subsequent pooling layer, which condenses the spatial dimensions while preserving the most prominent information. Preserving this information across a continuous stack of convolutional and pooling layers allows the network to progressively map low-dimensional input data into high-dimensional, abstract features. At the base of this feature mapping process, the initial layers capture simple, low-level spatial patterns, such as discrete edges or sharp facies boundaries. By mathematically combining these simple low-level edges in deeper layers, the network learns to recognize highly complex, high-level geological architectures, such as the overall sinuosity of a fluvial channel. Learning these complex high-level architectures is computationally efficient because the exact same kernel is applied universally across the entire grid, creating an inherent advantage known as translation invariance. This translation invariance ensures that if the network learns to identify a channel structure in one corner of the reservoir, it can accurately recognize that same structure anywhere else in the model, drastically reducing computational overhead and preventing overfitting. Further preventing this overfitting requires another critical component of the DL workflow: the strict use of validation and testing protocols. These validation protocols ensure that the hierarchical patterns learned by the network are robust and safely applicable to unseen data. Applying these robust, learned spatial patterns to unseen subsurface data forms the fundamental basis of modern **Deep Learning (DL)**-driven geomodelling workflows.

2.3.2. Geomodelling with Deep Learning

Modern DL-driven geomodelling workflows have evolved significantly, transitioning from simple pixel-based interpolations to the generation of highly sophisticated spatial representations. Generating these sophisticated representations relies on the ability of **DL** to directly learn complex geological features from large datasets or process-based simulations. Learning directly from such simulations bridges the historical gap between simplistic descriptive statistics and true geological realism (Laloy et al., 2017). To achieve this realism, **CNNs** have proven particularly successful in capturing the complex, non-stationary architectures of fluvial channels and facies distributions (Sun et al., 2023). Capturing these intricate architectures enables the rapid generation of stochastic realizations that successfully honor both the global geological concept and local, sparse conditioning data. Honoring these dual constraints while generating massive ensembles of stochastic realizations fundamentally enhances subsurface uncertainty characterization (Song et al., 2023). This unique capacity for rapid, constrained generation is primarily driven by specialized neural network frameworks, most notably generative architectures.

These generative architectures that are using **CNNs**, are best known for their ability to generate realistic images. The most-well known models include **VAEs**, **GANs** and Latent Diffusion Models. Though they are all different in terms of architecture, some core concepts remain similar. Generative models are designed to learn and approximate the complex joint probability distribution of the training data whereby elements are generated based on previously generated values (**he2025generative**). Moreover, these models use a latent space, designed to either store the encoded training data (**VAEs**) or be used to scale to an image through various upsampling steps (**GANs** and Latent Diffusion Models) (Prince, 2023).

2.3.3. Generative Adversarial Networks

Generative architectures, specifically Generative Adversarial Networks (**GANs**), achieve complex spatial generation through a competitive framework between two distinct neural networks (Goodfellow et al., 2014). These two networks are defined as a Generator (**G**) and a Discriminator (**D**). The generator is tasked with creating synthetic spatial samples from random noise, while the discriminator is simultaneously tasked with distinguishing these synthetic samples from actual real-world training data. The continuous competition to outsmart one another drives a minimax game, which is mathematically expressed as:

$$\min_{\mathbf{G}} \max_{\mathbf{D}} V(\mathbf{D}, \mathbf{G}) = \mathbb{E}_{\mathbf{x} \sim p_{\text{data}}(\mathbf{x})} [\log \mathbf{D}(\mathbf{x})] + \mathbb{E}_{\mathbf{z} \sim p_{\mathbf{z}}(\mathbf{z})} [\log(1 - \mathbf{D}(\mathbf{G}(\mathbf{z})))] \quad (2.1)$$

In this equation, $\mathbf{x}$ represents the real data, and $\mathbf{z}$ represents a random latent vector sampled from a prior distribution $p_{\mathbf{z}}$. The first term represents the discriminator successfully recognizing real geological data, while the second term represents the discriminator successfully catching fake data generated by $\mathbf{G}(\mathbf{z})$. During training, the discriminator seeks to maximize this overall value function to perfect its detection abilities, while the generator seeks to minimize it to successfully fool the discriminator. An overview of these steps, describing a single iteration of the model, is shown in figure B.2. By suc-

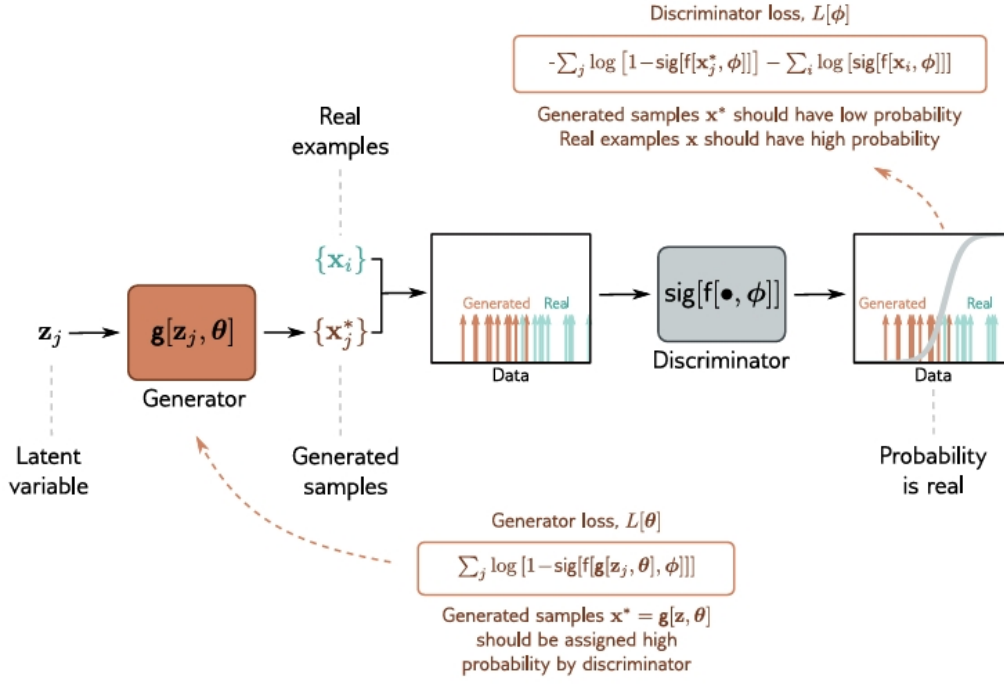


Figure 2.5: Basic overview of a standard GAN architecture with a Generator, constructing fake samples from a random latent variable z_j and a Discriminator deciding which of the two samples is the real sample (Prince, 2023).

Successfully fooling the discriminator, the generator ultimately learns to produce high-fidelity geological realizations without requiring any explicit, hard-coded spatial rules.

GAN Inversion

Figure B.2 illustrates the generation of high-fidelity, unconstrained realizations, which is a powerful capability; however, practical reservoir modelling requires these models to honor sparse physical constraints like hard well data. Hard well data constraints are typically honored through optimization techniques (Bhavsar et al., 2024; Chan & Elsheikh, 2019; Rongier & Peeters, 2025c). Several techniques have been proposed including applying inference networks (Bhavsar et al., 2024; Chan & Elsheikh, 2019), Gaussian Mixture modelling (Bhavsar et al., 2024), latent space optimization and inversion (Rongier & Peeters, 2025c). Where inference and GMM (ADD EXPLANATION ON THESE TECHNIQUES) GAN inversion operates strictly within the model's latent space after the training phase is complete. Completing the training phase allows the algorithm to iteratively search and optimize this latent space to identify a specific latent vector z . This specific latent vector z generates a 3D volume perfectly matching the sparse well data. Matching the sparse well data securely conditions the generative model to real-world subsurface observations. Moreover, a second proposed technique by Rongier and Peeters (2025c) is pivotal tuning, a technique originally designed to fine-tune the generator to comply a set latent space (Roich et al., 2022).

3 Methodology

This chapter describes the project workflow, beginning with the generation of 3D synthetic datasets used for training, validating, and testing the models (Section 3.1). Section 3.2 details the GAN framework, including the network architectures, the facies representation strategy, and the loss functions employed during training. Finally, Section 3.3 introduces the metrics and simulation methods used to evaluate the geological realism and practical utility of the generated realizations.

3.1. Dataset Generation

In order to test the applicability of GANs several new datasets had to be generated. Training deep learning algorithms for 3D geomodelling requires a substantial volume of high-fidelity training data. The new datasets were generated using FLUMY (Martin et al., 2016), a process-based stochastic modelling software designed to reproduce meandering channelized reservoirs. Based on sedimentological rules, the software reproduces facies type, grain size and deposition time for each voxel in the 3D volume.

3.1.1. Dataset Descriptions

To test the GAN in different scenarios, three distinct geological conceptual models were curated to capture a range of environments for the Delft Sandstone reservoir. The setting specific input parameters are shown in Table 3.1. The hypothesized depositional environments are split across three discrete

Parameter	Setting 1	Setting 2	Setting 3
Net-to-Gross (%)	67	67	67
Max. Channel Depth (m)	5	6	7
Max. Sand Body Extension	80	100	120

Table 3.1: NEXUS input parameters for the three training datasets.

suites, Ranging from a Lower plain to Upper plain deltaic characteristics in terms of increasing channel depth and wider sand bodies. Beyond these constraints, all generated models share a uniform reservoir scale with a total grid resolution fixed at $32 \times 128 \times 128$ voxels. In order to allow for well placement in the samples, the final dimension should encompass an area of at least 2000×2000 m as the **DEL-GT-01** and **DEL-GT-02-S2** have an 1500m spacing between them. Establishing this specific grid resolution combined with a standard voxel dimension of $20 \text{ m} \times 20 \text{ m} \times 1 \text{ m}$ means that the total simulated domain covers a volume of $2560 \text{ m} \times 2560 \text{ m} \times 32 \text{ m}$. Every single voxel within this acute 32 m vertical interval is ultimately encoded with 1 of the 14 distinct facies (see ??) available in FLUMY, of which only 9 facies actually occurred in the sample ensemble due to the settings specific parameters.

3.1.2. Sample preprocessing

To comprehensively assess the capabilities of GANs, the synthetic training datasets are prepared at two distinct categorical resolutions. These two distinct resolutions consist of a high-fidelity 9-facies dataset and an upscaled 3-facies dataset. The high-fidelity 9-facies dataset preserves the detailed depositional environments generated by FLUMY, serving as a baseline to evaluate the model's pure generative capacity and its ability to replicate highly complex spatial distributions. While evaluating this generative capacity is crucial, practical field application requires conditioning the model to actual subsurface observations. These actual subsurface observations, derived from the cored section of the **DEL-GT-01** well, are strictly classified into three primary facies associations (Section 2.1.1). Because the hard well data only resolves to three distinct associations, the original FLUMY depositional environments must be grouped accordingly to create the upscaled 3-facies dataset. This grouping process aggregates the detailed environments into broader categories, as detailed in Table 3.2.

Value	Facies Key	Color	Description
1	Channel & Point Bar (FA1)		Grouped proximal deposits: Channel Lag (1), Point Bar (2), Sand Plug (3), and Crevasse Splay I (4).
2	Levee & Crevasse (FA2)		Grouped intermediate deposits: Crevasse Channel (5), Crevasse Splay II (6), and Levee (7).
3	Overbank Fines (FA3)		Grouped distal deposits: Overbank (8), Mud Plug (9), Hemipelagic Plug (10), Wetland/Paleosol (11), Draping (12), and Pelagic (13).

Table 3.2: Overview of the upscaled 3-facies grouping used for the generative modelling, aggregated from the original 13 FLUMY depositional environments (See Table ?? for the original facies).

Beyond this grouping process, the categorical nature of both the 3-facies and 9-facies datasets necessitates a final preprocessing step known as one-hot encoding (Goodfellow et al., 2016). One-hot encoding is critical because standard neural networks process continuous numerical gradients and might otherwise incorrectly interpret integer facies labels as having an ordinal, mathematical relationship. This ordinal misinterpretation is prevented by transforming the original 3D volume into an expanded 4D tensor, where class is assigned its own dedicated spatial channel. Each spatial channel indicates the presence or absence of a specific facies using values of 1 and -1 (Read more on one-hot encoding appendix in B.1).

3.2. Fluvgan Framework

3.2.1. Model Architecture

The GAN architecture used in this project is based on the *Fluvial Generative Adversarial Network (FluvGan)* model, introduced by Rongier and Peeters (2025a). This architecture adapts the BigGAN framework (Brock et al., 2019) to manage the complex connectivity and high resolution necessary for realistic reservoir models. Where Rongier and Peeters (2025b) used a 2-channel approach for their model, one for grain size and one for deposition time, the model used in this project comprises of either 3 or 9 channels, where each channel represents another facies based on the one-hot encoding strategy.

The dynamic channel depth is accommodated by an architecture comprising a Generator (**G**) and a Discriminator (**D**) built with 3D Residual blocks. These blocks enable the generator to transform a, in this case, 100-dimensional latent vector $\mathbf{z}^*$ into a $C \times 32 \times 128 \times 128$ volume. This volume begins as a $512 \times 4 \times 4 \times 4$ feature map, created via a 3D transposed convolution, and is then expanded by five residual upscaling blocks. The final spatial resolution is processed through a Tanh activation, ensuring the C facies channels are scaled from -1 to 1. Finally, an argmax function scales the one hot encoded tensor back to integers, ensuring that a $32 \times 128 \times 128$ volume inhabited by class numbers remains. Every generated voxel is then evaluated by the discriminator, acting as a binary classifier to reduce the $C \times 32 \times 128 \times 128$ input to a single scalar logit. This logit is produced through five residual downscaling blocks that iteratively reduce spatial resolution while increasing the channel count to 512. A simple visual representation of the model can be seen in figure 3.1. In total, the generator involves roughly 11.6 million trainable parameters while the discriminator has about 2.0 million trainable parameters.

Evaluating the generated realizations begins by analyzing the foundational mathematical metrics, specifically Binary Cross-Entropy (BCE). BCE is deployed because the generator processes one-hot encoded facies, effectively turning the spatial generation into a *multi-channel binary classification task*. This classification task relies on BCE to penalize pixel-wise deviations between the generated probabilities ($\hat{y}_i$) and the true categorical labels (y_i), defined mathematically across N voxels as:

$$\mathcal{L}_{BCE} = -\frac{1}{N} \sum_{i=1}^N [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)]$$

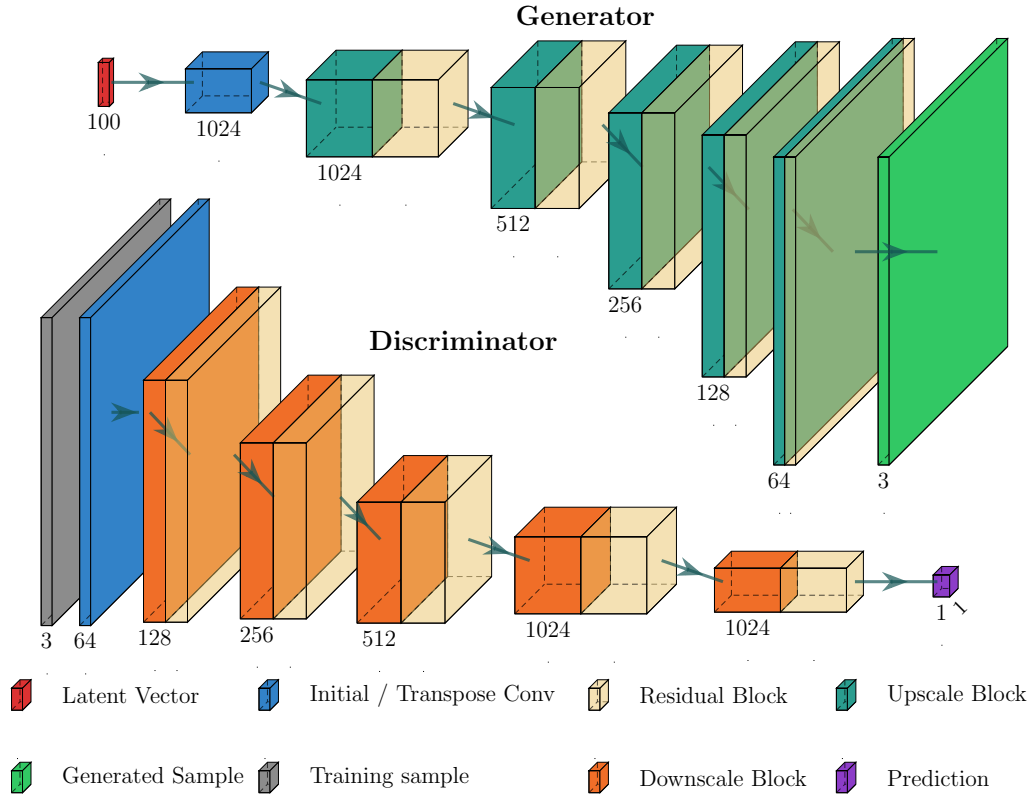


Figure 3.1: **Top**) The Generator initiates from a 100-dimensional latent vector and expands spatial volume through successive upscale and residual blocks. During this process, feature channel density is progressively reduced from 1024 down to 3 channels to construct the final generated volume. **Bottom**) The Discriminator mirrors this pathway by receiving a 3-channel sample (training or generated) and contracting spatial dimensions through downscale blocks. To retain high-level structural patterns during downsampling, feature channels are steadily increased from 3 up to 1024 before compressing into a single purple block representing the final prediction scalar.

During adversarial training, instability often manifests as exploding logarithmic losses when either the Generator **G** or Discriminator **D** becomes too confident in its predictions. Furthermore, poorly calibrated networks are highly susceptible to severe mode collapse, where the generator's stochastic capability is restricted. To investigate the optimal model architecture that yields stable training and diverse geological outputs, a systematic hyperparameter tuning study was conducted.

Hyperparameters

Hyperparameter tuning involves testing various combinations of network settings to evaluate their impact on model performance. The primary objective is to identify an optimal configuration that minimizes the discrepancy between the generated models and the input data, while actively preserving spatial ensemble diversity. To achieve this, baseline parameters (such as the optimizer and batch size) were kept constant (see table 3.3), while a specific set of architectural and training hyperparameters were varied stepwise. The complete search space for this hyperparameter study is outlined in Table 3.4.

Hyperparameter	Fixed Value	Description
Optimizer	Adam ($\beta_1 = 0, \beta_2 = 0.999$)	Standard optimization algorithm.
Batch Size	64	Number of samples processed per iteration.
Kernel Size	$3 \times 3 \times 3$	Dimensions of the 3D Convolutional filters.

Table 3.3: Fixed baseline configurations for the Fluvgan architecture optimization study.

Hyperparameter	Search Space / Value	Description
Learning Rate G (α_G)	10^{-3} , 10^{-4} , Dynamic	Generator step size during gradient descent.
Learning Rate D (α_D)	3×10^{-3} , 10^{-2} , Dynamic	Discriminator step size; testing a two-timescale update rule.
num iter D	1, 2	Number of consecutive Discriminator updates per Generator update.
Discriminator Penalty	None, 1, 16, 32	Gradient penalty applied to stabilize Discriminator confidence.
Double Conv Blocks	On, Off	Architectural toggle for adding secondary convolutional layers.
Double ResBlocks	On, Off	Architectural toggle for adding secondary residual blocks.
Training Samples	10,000, 20,000	Total volume of synthetic baseline data fed to the network.

Table 3.4: Hyperparameter search space for the Fluvgan architecture optimization study.

Among the evaluated parameters, the learning rates (α_G and α_D) along with the Discriminator update frequency (*num iter D*) are the most critical, as they directly dictate the competitive equilibrium of the adversarial minimax game. Tuning these factors enables the evaluation of the Two-Timescale Update Rule (TTUR) to maintain stable convergence and prevent mode collapse. Furthermore, the Discriminator Penalty serves as a vital regularizer to constrain gradient norms, ensuring the discriminator’s confidence does not overpower the generator early in training. Lastly, the architectural toggles (Double Conv and Residual Blocks) paired with data scaling systematically isolate whether network capacity or dataset volume is primary in preserving the necessary spatial and structural diversity of the generated fluvial systems.

3.2.2. Optimization & Inversion

While the 9-facies models are utilized to evaluate unconstrained generative performance, the 3-facies models are deployed to test practical conditioning against the **DEL-GT-01** well data. To successfully condition this generative model to real-world observations, the inversion workflow (introduced by (Rongier & Peeters, 2025c)) relies on a two-step mechanism: Latent Space Optimization and Pivotal Tuning (Roich et al., 2022).

First, Latent Space Optimization searches for the optimal random noise vector that best approximates the well data while keeping the pre-trained generator’s weights frozen. Because this often provides a close structural approximation but rarely a perfect match, it is followed by Pivotal Tuning. This second phase uses the optimized noise vector as a static anchor and temporarily unfreezes the generator’s internal weights to make microscopic adjustments. This forcefully aligns the localized output with the exact well data, while regularization prevents the collapse of broader geological realism. (A detailed mathematical formulation of this inversion process is provided in Appendix C.1).

3.3. Evaluation Metrics

Evaluating the performance of generative models on 3D spatial grids presents significant challenges, as standard metrics like Binary Cross-Entropy (BCE) or direct voxel-by-voxel assessments fail on unconditioned models. Therefore, the evaluation framework relies on metrics that capture complex geological structures and spatial uncertainty across an ensemble of generated samples.

To evaluate how closely the generated realizations resemble the training data’s global morphological structures, we employ the Mean Sliced Wasserstein Distance (MSWD). Based on the optimal transport principles of the Wasserstein distance (Panaretos & Zemel, 2019), which was highlighted for geological 3D structures by Rongier and Peeters (2025b), MSWD evaluates structural similarity without requiring rigid pixel alignment. By projecting 3D spatial distributions onto 1D linear slices and averaging the

results, MSWD efficiently bypasses the computational bottleneck of high-dimensional grid volumes. It serves as our primary indicator of global architectural consistency and mode collapse during the training process. During training, the model saves 10% of its training data exclusively for evaluation used the MSWD metric, ensuring the samples are not passed on into the Discriminator.

Beyond evaluating structural realism, quantifying the spatial uncertainty and generation diversity across the ensemble is critical. To achieve this, pixel-wise Normalized Shannon’s Entropy (Shannon, 1948) is utilized. By calculating the entropy at each specific spatial coordinate across multiple generated models, this metric serves as a direct validation of the model’s diversity. High entropy values at a given pixel indicate that the generator is capable of producing a varied distribution of facies at that location, whereas low entropy exposes areas of rigid uniformity or potential mode collapse. This pixel-wise evaluation visually maps regions of spatial conflict and agreement, which is then aggregated into a definitive scalar measure of spatial fidelity using the Jensen-Shannon (JS) divergence (Lin, 1991). As a symmetric metric based on Kullback-Leibler divergence (Kullback & Leibler, 1951), the JS divergence measures the overall distributional disagreement between the baseline flumy and GAN models. (The full mathematical derivations for these evaluation metrics are detailed in Appendix C.2).

Data conditioning

While JS divergence defines the overall ensemble spread, after optimization and inversion, each individual GAN model within that ensemble must still strictly honor the physical reality at the wellbore. This physical reality at the wellbore acts as a strict spatial constraint, the localized mismatch of which is calculated using the Macro F1-score. The Macro F1-score calculates the unweighted average of the harmonic mean of precision and recall across all C individual facies classes:

$$F1_{macro} = \frac{1}{C} \sum_{c=1}^C \frac{2 \cdot \text{Precision}_c \cdot \text{Recall}_c}{\text{Precision}_c + \text{Recall}_c}$$

Where for each class the precision and recall scores have to calculated:

$$\text{Precision} = \frac{TP}{TP + FN}$$

$$\text{Recall} = \frac{TP}{TP + FN}$$

where true positive (TP) represents the number of correctly predicted positive pixels, false positive (FP) the predicted positive but actually negative pixels, false negative (FN) the predicted negative but actually positive pixels.

Evaluating every facies class independently provides a highly robust performance metric that heavily penalizes the model if it ignores underrepresented minority classes (i.e. a fair metric when dealing with class imbalance). These underrepresented minority classes would otherwise be completely overshadowed by the more common classes if standard statistical accuracy was deployed to assess the well conditioning.

4 Results

The presentation of the results will follow the order of the workflow: First of all the curated datasets will be displayed with the addition of statistical parameters that can be used to evaluate the difference between the samples in the datasets and the synthetic GAN samples. Secondly, the post training results are presented which presents the results of the hyperparameter tuning, the cGAN unconstrained performance to grouped facies classes as well as the cGAN unconstrained performance when presented with all facies classes. Finally, the results of the optimization and inversion results are presented, after the model was constrained to the Delft Geothermal Reservoir well data.

4.1. Datasets

In total, 20,000 samples were generated for the three datasets as defined in table 3.1. These generated datasets are visually summarized in figure 4.1: a 2D MDS plot clearly shows a grouping effect per setting, indicating the samples of different settings do in fact display varying higher-dimensional differences. This can be visually confirmed by the two random 3D facies plots per setting, showing that a lower sand body extension as input parameter yields samples with thinner sandbodies and more erratic meandering behavior in samples of this size. The right most plots show the pixel-wise entropy computed over an ensemble of 1000 samples per setting, setting a baseline for the entropy readings we look to achieve from a diverse synthetic dataset produced by the trained GAN model. This threshold configuration can be further evaluated by analyzing table 4.1, which provides an exhaustive overview of the global facies distributions across the ensemble. As can be observed in this overview, the structural

Facies	Setting 1 (%)	Setting 2 (%)	Setting 3 (%)
1: Channel Lag	1.6	1.8	1.8
2: Point Bar	67.5	66.3	65.4
3: Sand Plug	1.8	1.6	1.4
4: Crevasse Splay Core	0.0	0.0	0.0
5: Crevasse Channel	0.1	0.1	0.1
6: Crevasse Splay Delta	0.0	0.0	0.0
7: Levee	5.1	6.5	7.7
8: Overbank	10.2	11.4	11.9
9: Mud Plug	13.6	12.4	11.5

Table 4.1: Global categorical facies distribution profiles compared across the three synthetic training datasets for all available classes.

facies, hereafter referred to as classes, exhibit a severe statistical imbalance throughout the data volume. This severe imbalance is a result of dominant classes occupying over 60% of the total grid volume, while corresponding minority classes are entirely absent due to the fundamental boundary limitations of the FLUMY NEXUS parameters. The volumetric presence of classes 4, 5, and 6 is particularly suppressed, yielding an exceptionally low occurrence rate ranging between 0.0% and 0.1%. Although the classes will be grouped to form three classes instead of nine, the class imbalance will remain with the second class only occurring around 5|5 of the time while the first class occurs approximately 70% of the time. Such depressed occurrence rates could present an optimization bottleneck as the random occurrence of these volumes throughout the samples could confuse the model into finding an optimized setting that totally disregards these classes. The Discriminator could get stuck on deciding whether a sample is real or fake by judging mostly based on class 1, which leads to the Generator trying to improve the structure of this class, completely disregarding the other classes. Moreover, the high class imbalance in the data has to

be accounted for when evaluating the model, especially with regards to the well mismatch evaluation.

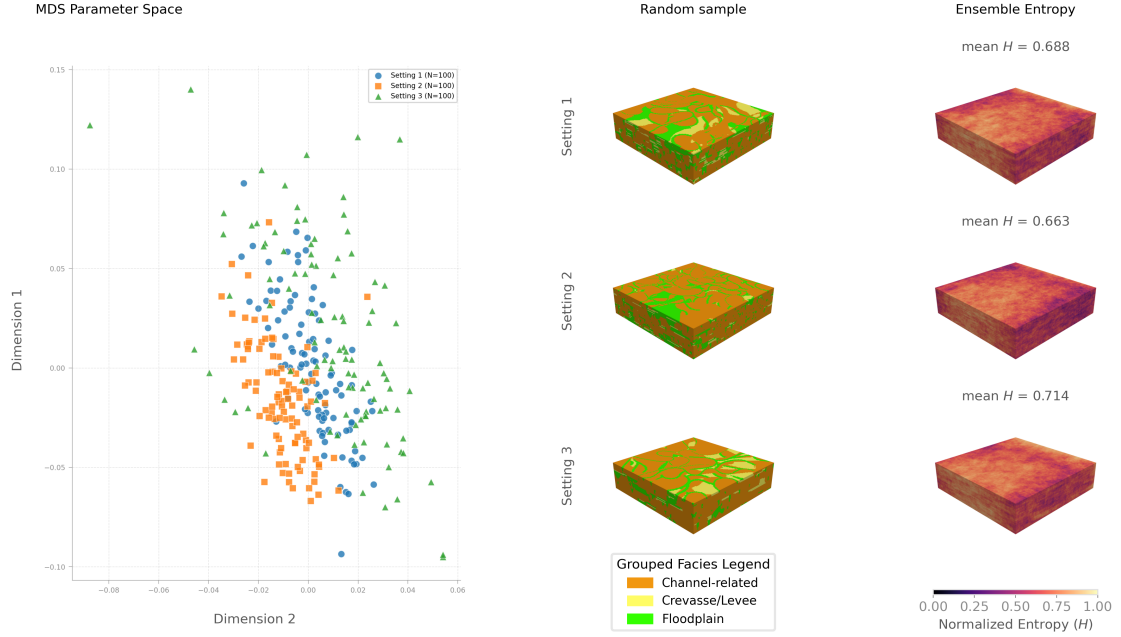


Figure 4.1: MDS plot with two 3D facies samples displayed and a 3D pixel-wise entropy per setting.

4.2. Post Training Generative Results

All the models were trained on 2 NVIDIA-A100 GPU's with each 80GB of unified memory. When trained for 25 epochs on 20000 samples, the execution time took approximately 3-4 hours.

4.2.1. Hyperparameter Tuning

Figure 4.2 and table 4.2 provide an overview of the hyperparameter tuning study. All trained model reserved 10% of their training data exclusively for validation: computing the [Sliced Wasserstein Distance \(SWD\)](#) metric for both training and validation datasets. Figure 4.2 includes tracking of the Sliced Wasserstein Distance (SWD) metric across 25 epochs, a randomly generated sample by the trained model and a 3D entropy plot of an ensemble of 1000 [GAN](#) generated samples. Complementary, the Jensen-Shannon Divergence (JSD) and Mean SWD which both compare the ensemble of 1000 generated samples per run to a test set of 1000 samples is provided in table 4.2. Based on the metrics from

Run	Epochs	disc penalty	double conv	double resblock	JSD ↓	MS-SWD ↓	mean H_{norm} ↑
1	25	no	off	off	0.0718 ± 0.0157	0.08865	
2	25	no	on	off	0.1161 ± 0.0243	0.06842	
3	25	no	on	on	0.0458 ± 0.0076	0.08271	
4	25	32	on	on	0.0612 ± 0.0314	0.09224	
5	25	16	on	on	0.0155 ± 0.0067	0.07986	
6	25	1	on	on	0.0202 ± 0.0038	0.07690	
7	25	16	on	on	0.1190 ± 0.0324	0.11924	
8	25	16	on	on	0.0278 ± 0.0045	0.13389	

Table 4.2: Overview of hyperparameters and evaluation metrics of each run. Run 5, based on architecture 4 by (Rongier & Peeters, 2025b) yields the best JSD and second to best MS-SWD value.

table 4.2 and entropy plots from figure 4.2, the best model was achieved in run 5 with a JSD of 0.0155, MS-SWD of 0.07986 and with the best pixel-wise entropy value out of all the runs (0.79), supported by the 3D entropy plot which shows almost no structure visible in any region. The MS-SWD metric

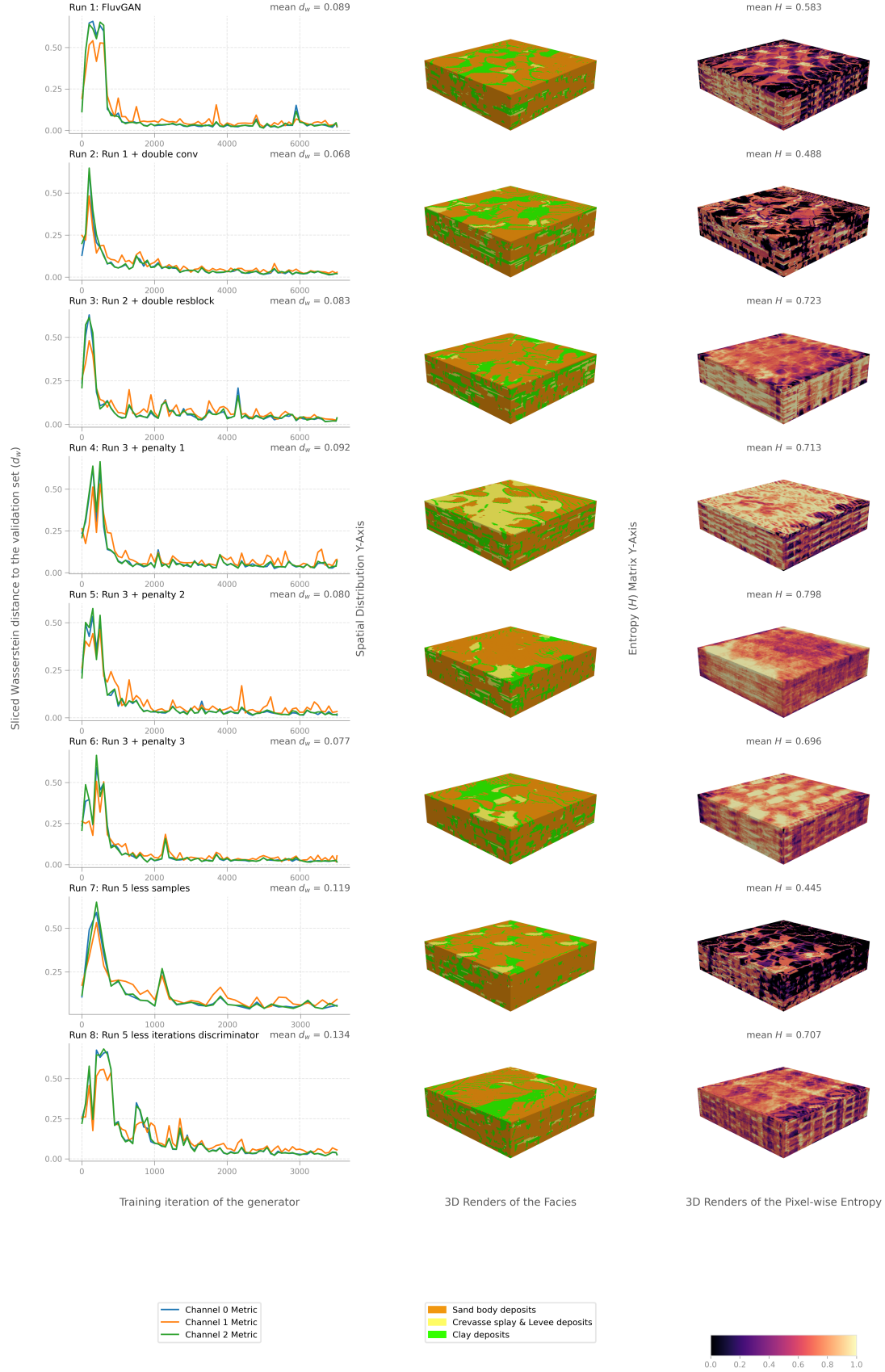


Figure 4.2: Ablation study starting from FluvGAN and progressively adding/adjusting parameters to observe their effect on training, geological plausibility and sample diversity. *Run 1*: Standard FluvGAN architecture from (Rongier & Peeters, 2025b). *Run 2*: Addition of double convolutional layers. *Run 3*: Addition of double residual blocks (i.e. skip connections) on top of double convolutions. *Run 4*: Addition of a discriminator penalty every 32 iterations. *Run 5*: Changed discriminator penalty to every 16 iterations. *Run 6*: Changed discriminator penalty to every iteration. *Run 7*: Run 5 hyperparameters but decreased to 10000 training samples. *Run 8*: Run 5 hyperparameters but changed number of iteration of the discriminator to 2.

showed all samples decreasing in Sliced Wasserstein's Distance, indicating a better fit to the training dataset. An interesting observation with regards to the **SWD** evolution over 25 epochs can be made: All plots show a stabilizing SWD after 2000 training iterations. From a **SWD** metric standpoint, this indicates that the model has stopped improving its samples compared to the validation dataset.

Moreover, the base **FluvGan** architecture lead to significant mode collapse looking at figure 4.2. Only after increasing the model depth (double convolutions & double residual blocks) did the model train properly without mode collapse. Moreover, when varying the discriminator R1 Regularization, applying the penalty every 16 iterations yielded the best result. All other option (no penalty, every 32 iterations and every iteration) all resulted in mode collapse. Finally, run 7 and 8 investigated whether decreasing the number of training samples from 20000 to 10000 and only updating the discriminator every 2nd iteration, yielded any improved results. Where the results from run 8 yielded similar results to run 6, the metrics from run 8 yielded a much higher mean JSD of 0.1190 ± 0.0324 . Additionally, looking at the entropy plot of run 8 in figure 4.2, The plot shows a significantly higher percentage of regions with low entropy values, indicating that most samples produced by the generator after training have similar pixel-wise values.

4.2.2. cGAN Unconstrained Performance

The model with run 5 parameters, of which a complete list with all exact parameters can be found in (add table to the appendix), was run for 100 epochs as well to see whether the results could improve. Although the the JSD and MS-SWD values did not vary significantly, the geological plausibility of the samples seems to increase significantly, suggested by figure 4.3: when visually, comparing the evolution from 25 to 100 epochs, especially the continuity of the clay deposits, the continuity of this facies seems to have increased (add comparison figure between these two samples! -> 20 3s samples from 25 epochs and 20 samples from 100 epochs.). This is further supported by the blob mean blob size calculation (see table) As already noted in section ??, the **SWD** metric evolution stops improving after around 2000 training iterations, suggesting that the model has stopped improving. However, when assessing the samples from a visual perspective, ... ADD EVOLUTION...

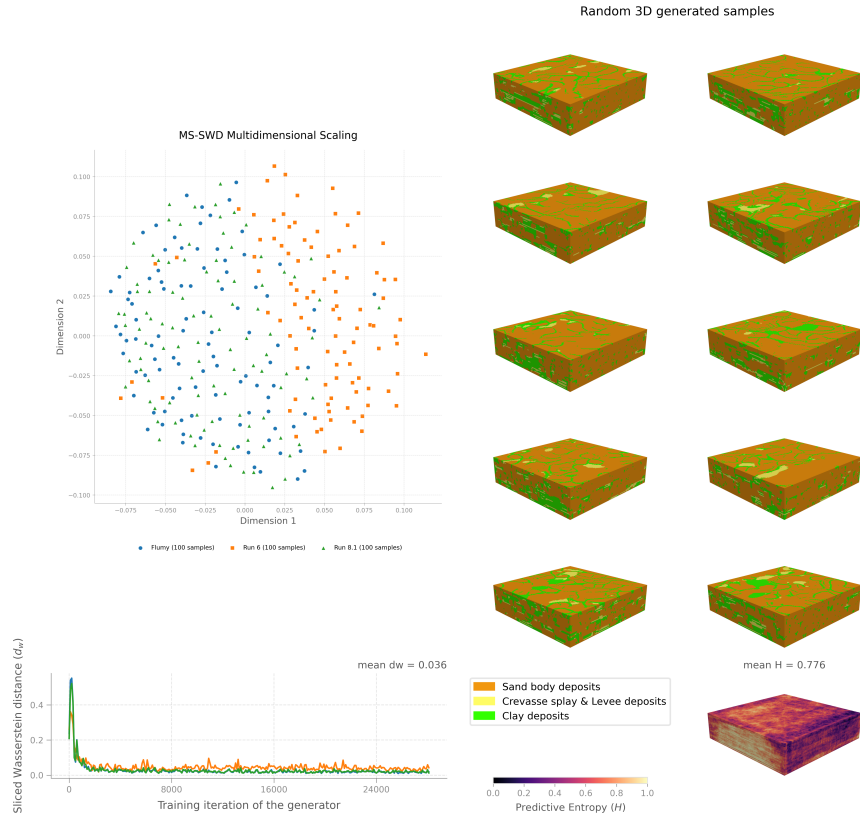


Figure 4.3: Overview of model trained with run 5 settings as described in table 4.2 and figure 4.2 for 100 epochs. The MDS plot shows a 2D representation of a high dimensional structure computed with the Wasserstein distance. For reference have 100 samples from run 6 been added, showing that those samples do not overlap with the training dataset samples, suggesting they lack in full complexity.

4.2.3. High-Fidelity 9-Class Generative Performance

Besides only evaluating the model trained on the grouped facies, the training was also conducted on the high fidelity case: testing whether reverting to 9 facies classes lead to significantly different results in terms of geological plausibility and sample diversity. Since the classes are highly unbalanced, this run tests what the capability limits of this model is. The model was trained for 25 epochs on 20000 samples, with the same hyperparameters as run 5. The results are summarized in figure 4.4. The **SWD** evolution indicates that the models ability to represent class structures, varies significantly per class. For channels 0 to 2 and 6 to 8 the **SWD** metric decreases immediatly from the start of training, indicating that the model is able to represent these classes well. However, for channels 3 ad 5, the **SWD** metric no improvement is made, indicating that the model is not able to represent these classes well. Channel 4 decreases over the full amount of iterations with the possibility to decrease even more. Finally, channel 0, channel 2 and channel six, which had a low **SWD** from the start onwards, display occasional increases in the metric.

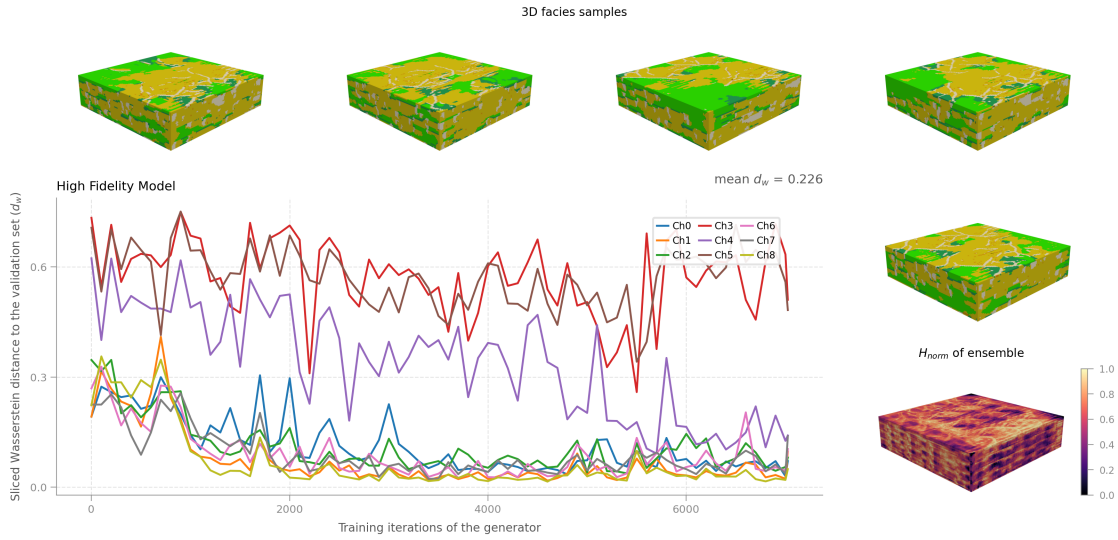


Figure 4.4: Overview of (best) high fidelity model trained over 25 epochs.

From a visual perspective, the samples show signs of being able to recreate geological features. This claim is supported by the **Multi-Dimensional Scaling (MDS)** plot (ADD Figure to appendix!) and the corresponding 3D facies plots. However, when evaluating the pixel-wise entropy plot we can observe that the samples show a significant amount of structure, indicating that the model is not able to represent the full complexity of the training dataset. This is further supported by the mean pixel-wise entropy value of ??, which is significantly lower than the mean pixel-wise entropy value of 0.79 for the grouped facies case.

4.3. Post-Inversion Results

During optimization and inversion, the tuned model was conditioned to the well data presented in Section 2.1. The evaluation metrics were computed exclusively on the first 32 meters of well data from DEL-GT-01 and DEL-GT-02-S2, yielding a total of 64 spatial conditioning points for the Macro F1-score evaluation. During the initial Latent Space Optimization phase, 100 latent vectors were optimized to fit the well data. These outputs are henceforth referred to as “realizations”, as they represent models explicitly conditioned to real-world subsurface observations.

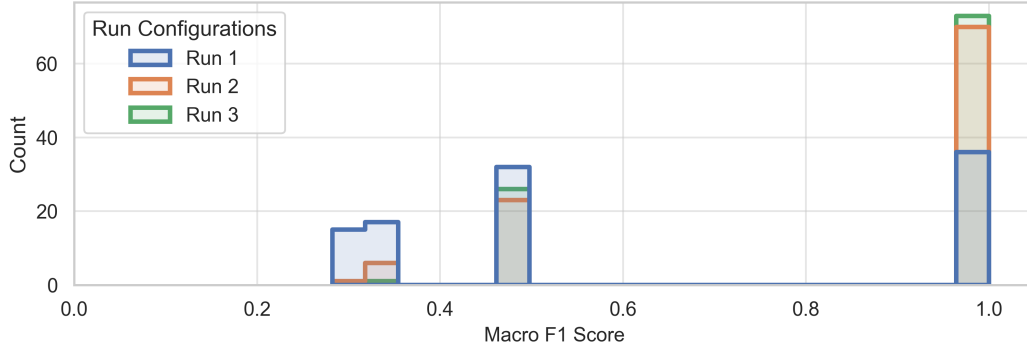


Figure 4.5

As illustrated in Figure 4.5, the percentage of realizations achieving a flawless fit with the conditioning points is highly dependent on the optimization threshold (Th). The threshold (Th) defines a specific spatial radius around the wellbore, measured in voxels, within which the generative model is penalized for mismatching the hard conditioning data. Specifically, voxels located beyond this distance threshold are assigned a weight of zero in the context loss function, ensuring that the inversion process enforces localized geological alignment without over-constraining the broader, unobserved reservoir architecture (See figure C.1). Table 4.3 details the Macro F1-scores and spatial entropy metrics achieved across these varying threshold configurations.

Run	Th	Macro F1 $\uparrow$	Mean $H_{norm,P}$ $\uparrow$	Mean $H_{norm,I}$ $\uparrow$	Count P $\uparrow$	Count I $\downarrow$
Run 1	1	0.618539	0.641889	0.664907	36	64
Run 2	2					
Run 3	5	0.835997	0.634802	0.629816	70	30
Run 4	10	0.861774	0.596361	0.582823	73	27
Run 5	15					

Table 4.3: Optimization metrics evaluated across different run configurations. The column ‘Th’ indicates the specific threshold value defining the spatial region affected by latent optimization. Subscript indices P (Perfect) denote the subset of generated samples achieving a flawless well alignment ($F1 = 1.0$), while I (Imperfect) refers to the remaining sample subset that did not fit perfectly ($F1 < 1.0$). Normalized spatial entropy values (H_{norm}) are provided respectively for both groups alongside categorical sample counts.

Concurrently, the distribution of realizations achieving a perfect fit exhibits a rapid, non-linear response to the threshold parameter. Moving from a threshold of 1 (Run 1) to a threshold of 5 (Run 3) yields a near-exponential increase in the number of perfectly conditioned realizations, jumping from 36 to 70. However, as the threshold continues to increase past this point, the rate of improvement plateaus, reaching only 73 perfect realizations at a threshold of 10. This plateau suggests a fundamental limit to how much a frozen generator can bend its learned prior to fit localized data, demonstrating why subsequent Pivotal Tuning is required to bridge the remaining spatial mismatch.

Furthermore, Table 4.3 highlights a critical trade-off between strict well conditioning and spatial diversity. As the optimization threshold increases, the normalized spatial entropy (H_{norm}) systematically decreases for both the perfectly aligned (P) and imperfectly aligned (I) subsets. This mathematical

drop in entropy indicates that enforcing stricter conditioning constraints forces the optimization algorithm to sample from a narrower, more restricted region of the latent space. Consequently, while the well data is honored more frequently, the overall stochastic diversity of the generated ensemble diminishes. Visual confirmation of this phenomenon is provided in Appendix ??, which demonstrates that the spatial extent of low-entropy, high-certainty regions expands proportionally with the applied threshold.

5 Discussion & Recommendations

5.1. Interpretations of key results

The implementation of the Fluvgan framework demonstrated highly stable training dynamics throughout the experimental phase. Once an optimal network architecture was established, a process that proved to be relatively straightforward, the Generator and Discriminator maintained a steady loss equilibrium (figure ??). This stability is particularly notable given the high-dimensional complexity of 3D spatial data, supporting the statement by (Rongier & Peeters, 2025b) that GANs are more stable than given credit for.

However, evaluating the unconstrained conditional GAN (cGAN) reveals a stark contrast between macro-scale replication and true spatial complexity. As demonstrated by the global volume statistics, the cGAN successfully mimics the overall facies distributions of the geological prior, producing structural proportions that appear nearly identical to the training data. Yet, spatial entropy metrics indicate that the model lacks a rigorous understanding of the true complexity inherent in the synthetic datasets. The severe drop in normalized spatial entropy mathematically confirms a state of mode collapse, wherein the network repeatedly generates similar spatial configurations rather than exploring the full architectural diversity of the fluvial-deltaic system.

Despite these generative limitations in unconstrained scenarios, the model excelled during the post-training conditioning phase. The latent-space optimization step already achieved an average macro-F1 score of 0.6 where realization either achieved 100% accuracy or $\pm 30\%$ accuracy. After Pivotal tuning of the Generator the mean F1 score increased slightly to ... In terms of practical use cases having some samples not honouring the well data is not a problem as the user can simply discard those samples and use the ones that do. The fact that the model is able to generate realizations that are 100% accurate in honouring the well data is a significant achievement, especially considering the complexity of the geological structures being modeled. Combined with the ability of GANs once trained to generate samples incredibly fast (1 second for a 3D sample of 32x128x128 voxels), shows its potential for real-time applications in subsurface modeling and reservoir characterization.

5.2. Limitations of the current work

Severe Class Imbalance: (Weighted Class Loss) The foundational FLUMY training datasets are characterized by extreme volumetric imbalance. Dominant channel structures overshadow intermediate facies (such as the Levee & Crevasse association), resulting in near-zero occurrence rates for specific minority classes. This statistical suppression inherently bottlenecks the inversion process, as the network struggles to accurately represent structures it rarely encountered during training.

Ensemble Mode Collapse: The cGAN's susceptibility to mode collapse fundamentally limits its utility for comprehensive uncertainty quantification. Because the model fails to capture the full stochastic diversity of the training data, the resulting ensemble underestimates the true geological uncertainty present between sparse data points.

Grid Resolution and Scaling Constraints: The fixed voxel resolution of 128x128x32 restricts the simulated domain to a localized 2560m x 2560m x 32m volume. While sufficient for demonstrating proof-of-concept well conditioning between two nearby wellbores, this grid size cannot encapsulate the entire Delft Sandstone reservoir, presenting computational challenges for full-field upscaling.

5.3. Recommendations for future work

Full-Field Upscaling and Dynamic Simulation: Research should expand beyond static geological evaluation by scaling the generated volumes to full-field dimensions using patch-based generation or multi-scale GANs. Furthermore, the 99% conditioned realizations should be subjected to dynamic fluid flow

simulations to quantify how the ML-generated architectures impact actual geothermal heat extraction and fluid connectivity compared to traditional geostatistical models.

6 Conclusion

6.1. Answers to Research Questions

6.2. Final Remarks

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A FLUMY dataset generation

Value	Facies Key	Color	Description
0	Background		Undefined
1	Channel Lag		Coarse active channel fill
2	Point Bar		Meandering lateral accretion
3	Sand Plug		Fine-grained channel abandonment
4	Crevasse Splay I		Proximal high-energy overbank
5	Crevasse Channel		Feeder for splay deposits
6	Crevasse Splay II		Feeder for splay deposits
7	Levee		Bordering channel ridges
8	Overbank		Vegetated floodplain silt
9	Mud plug		Organic-rich anoxic sediments
10	Hemipelagic plug		Fine-grained marine deposits consisting of clay and silt-sized grains
11	Wetland/Paleosol		Organic-rich anoxic sediments
12	Draping		Fine-grained sediments conforming to topography
13	Pelagic		Deep-marine fine-grained suspension settling

Table A.1: Overview of the 9 facies classes that populated the training dataset. Due to specific setting in the flumy default non expert user settings (see figure 3.1), only 9 facies populated the eventual datasets.

Lithofacies ID	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Lithofacies	Pelagic	Draping	Wetland	Hemipela. Plug	Mud Plug	Overbank	Levee	Crevasse Splay II	Crevasse Splay Channel	Crevasse Splay I	Sand Plug	Point Bar / LAPS	Channel Lag	Undefined
Abbreviation	PL	DR	WL	HP	MP	OB	LV	CS2	CCH	CS1	SP	PB	CL	UDF
Default Grain Size Class	1	1	1	2	2	3	6	7	8	9	9	10	13	0
Default ϕ Class	11:14	11:14	11:14	9:10	9:10	8	5	4	3	2	2	1	-2	NA

Grain size classes and lithofacies ranges:

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Undefined	Clay (11:14)	Clay (9:10)	Clay (8)	Silt (7)	Silt (6)	Silt (5)	Silt (4)	Very Fine Sand (3)	Fine Sand (2)	Medium Sand (1)	Coarse Sand (0)	Very Coarse Sand (-1)	Gravel (-2)	Pebble (-3:-5)	Cobble (-6:8)
0.0625	0.125	0.1875	0.25	0.3125	0.375	0.4375	0.5	0.5625	0.625	0.6875	0.75	0.8125	0.875	0.9375	1
	Pelagic	Mud Plug				Crevasse Splay II			Crevasse Splay Channel						
	Draping	Hemipelagic								Crevasse Splay I					
	Wetland	Plug								Point Bar / LAPS					
		Overbank				Levee							Channel Lag		
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15

Figure A.1: Overview of all possible facies defined in the flumy user guide (add citation) with the respecting range of grain sizes.

B FluvGAN

B.1. One Hot Encoding Strategy

The one-hot encoding strategy used in this project created n spatial channels based on the number of individual classes passed on to the model. Where traditional one-hot encoding strategies map the values in a channel to either 1 or 0, referring to the presence or absence of a specific class, the final *Tanh* activation function in the *FluvGAN* model maps values to either -1 or 1. Figure B.1 provides an overview of one hot encoding of a vector. Using one-hot encoding categorical independent data can be

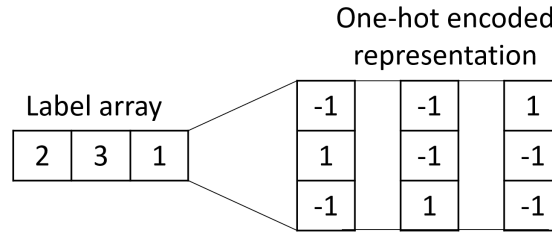


Figure B.1: Simple one-hot encoding representation by converting the original data array into a multitude of bipolar vectors. One-hot encoding can be applied to any type of categorical data.

correctly fed to the model, avoiding issues where the model thinks the numbers/values are correlated. The model may start to correlate the classes based on values, tending to overestimating the classes that do not occur often.

B.2. Fluvgan Architecture

Layer / Block Type	Output Shape	Param #
ConvBlockTranspose3d (Scale 1)	[1, 1024, 4, 4, 4]	6,553,600
DeepResBlockUpscale3d (Scale 2)	[1, 1024, 8, 8, 8]	4,066,816
DeepResBlock3d (Scale 2)	[1, 1024, 8, 8, 8]	4,066,816
DeepResBlockUpscale3d (Scale 3)	[1, 512, 16, 16, 16]	3,935,744
DeepResBlock3d (Scale 3)	[1, 512, 16, 16, 16]	1,017,600
DeepResBlockUpscale3d (Scale 4)	[1, 256, 32, 32, 32]	984,832
DeepResBlock3d (Scale 4)	[1, 256, 32, 32, 32]	254,848
DeepResBlockUpscale3d (Scale 5)	[1, 128, 32, 64, 64]	99,200
DeepResBlock3d (Scale 5)	[1, 128, 32, 64, 64]	27,072
DeepResBlockUpscale3d (Scale 6)	[1, 64, 32, 128, 128]	25,024
DeepResBlock3d (Scale 6)	[1, 64, 32, 128, 128]	6,880
BatchNorm3d + LeakyReLU	[1, 64, 32, 128, 128]	128
ConvBlock3d (Output)	[1, 9, 32, 128, 128]	15,552
Total Trainable Parameters		21,054,112

Table B.1: Detailed architectural specifications and layer parameters for the DeepGenerator3d network.

Layer / Block Type	Output Shape	Param #
ConvBlock3d (Input)	[1, 64, 32, 128, 128]	15,552
DeepResBlockDownscale3d (Scale 2)	[1, 128, 16, 64, 64]	20,992
DeepResBlock3d (Scale 2)	[1, 128, 16, 64, 64]	63,488
DeepResBlockDownscale3d (Scale 3)	[1, 256, 8, 32, 32]	83,968
DeepResBlock3d (Scale 3)	[1, 256, 8, 32, 32]	253,952
DeepResBlockDownscale3d (Scale 4)	[1, 512, 4, 16, 16]	335,872
DeepResBlock3d (Scale 4)	[1, 512, 4, 16, 16]	1,015,808
DeepResBlockDownscale3d (Scale 5)	[1, 1024, 4, 8, 8]	753,664
DeepResBlock3d (Scale 5)	[1, 1024, 4, 8, 8]	1,703,936
DeepResBlockDownscale3d (Scale 6)	[1, 1024, 4, 4, 4]	1,703,936
DeepResBlock3d (Scale 6)	[1, 1024, 4, 4, 4]	1,703,936
LeakyReLU	[1, 1024, 4, 4, 4]	0
LinearBlock (Output Prediction)	[1, 1]	1,025
Total Trainable Parameters		7,656,129

Table B.2: Detailed architectural specifications and layer parameters for the DeepDiscriminator3d network.

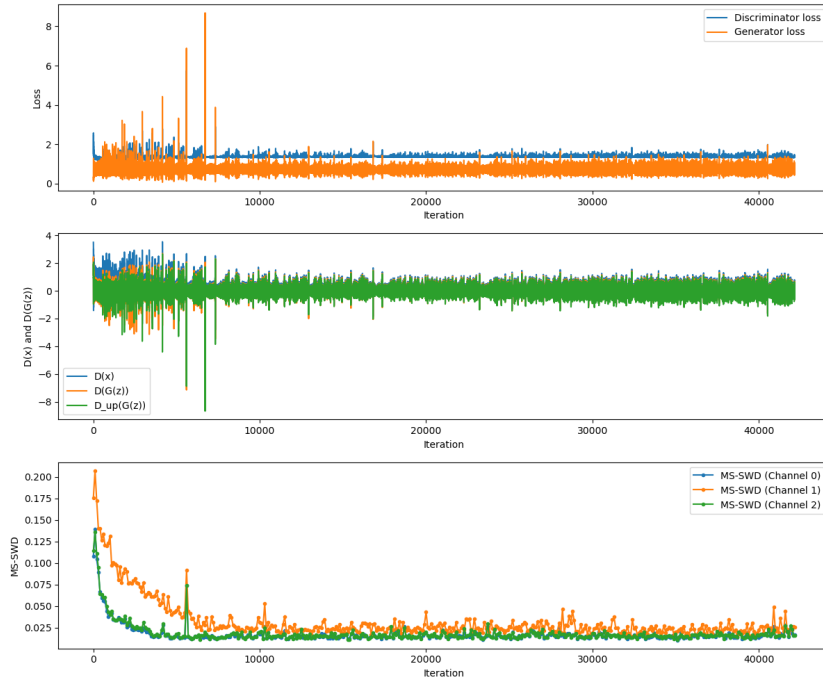


Figure B.2

C Mathematical Formulations for Optimization and Evaluation

C.1. Optimization & Inversion Framework

The two-step inversion process for conditioning the 3-facies generative models is mathematically defined by consecutive optimization phases: Latent Space Optimization followed by Pivotal Tuning.

C.1.1. Latent Space Optimization

During the initial Latent Space Optimization, the internal weights of the generator G are frozen. Gradient descent is utilized to iteratively search the latent space $\mathcal{Z}$ for an optimal latent noise vector $\mathbf{z}^*$ that minimizes the loss between the generated facies and the true geological observations.

Rather than evaluating the penalty exclusively at the exact well coordinates, a spatially weighted context mask is applied to enforce localized geological alignment without over-constraining the broader, unobserved reservoir architecture. A spatial influence radius is established around each well point, governed by a distance threshold (σ). This threshold acts as the standard deviation for a continuous Gaussian decay function, dictating the geostatistical “reach” of the well data.

Because fluvial-deltaic depositional systems are highly anisotropic—characterized by extensive horizontal continuity and limited vertical thickness—an anisotropic distance metric $d_A(v)$ is utilized, penalizing vertical deviations significantly more than horizontal ones. For any voxel v in the 3D grid $\mathcal{V}$, the spatial weight mask $M(v)$ is defined as:

$$M(v) = \begin{cases} \exp\left(-\frac{d_A(v)^2}{2\sigma^2}\right), & \text{if } d_A(v) \leq 3\sigma \\ 0, & \text{if } d_A(v) > 3\sigma \end{cases} \quad (\text{C.1})$$

While the threshold (σ) controls the smooth, continuous spread of the penalty, calculating gradients for distant voxels is computationally inefficient. Therefore, the mask is explicitly truncated to zero at a hard computational cutoff of 3σ , where the gradient contribution becomes mathematically negligible.

The objective function is thus defined as a spatially masked Context Loss, replacing the point-wise evaluation. It is evaluated over the grid volume $\mathcal{V}$ and the C total facies classes:

$$\mathbf{z}^* = \arg \min_{\mathbf{z}} \left[\sum_{v \in \mathcal{V}} \sum_{i=1}^C M(v) \cdot \mathcal{L}_{CE}(p_{v,i}, y_{v,i}) \right] \quad (\text{C.2})$$

where $\mathcal{L}_{CE}$ is the Categorical Cross-Entropy. Specifically, $y_{v,i}$ is the binary truth value (0 or 1) indicating whether class i is the correct broadcasted facies for voxel v (based on the nearest well data point), and $p_{v,i}$ is the generator’s predicted probability for that class.

Computational Scaling: Latent optimization is highly efficient. The computational cost scales linearly, $\mathcal{O}(N_w \cdot C)$, with the number of conditioning points N_w and facies classes C . Because the network weights are frozen, the gradient is only computed with respect to the latent vector $\mathbf{z}$ (typically of dimension 512 or 1024), making this phase fast and highly scalable to dense well configurations.

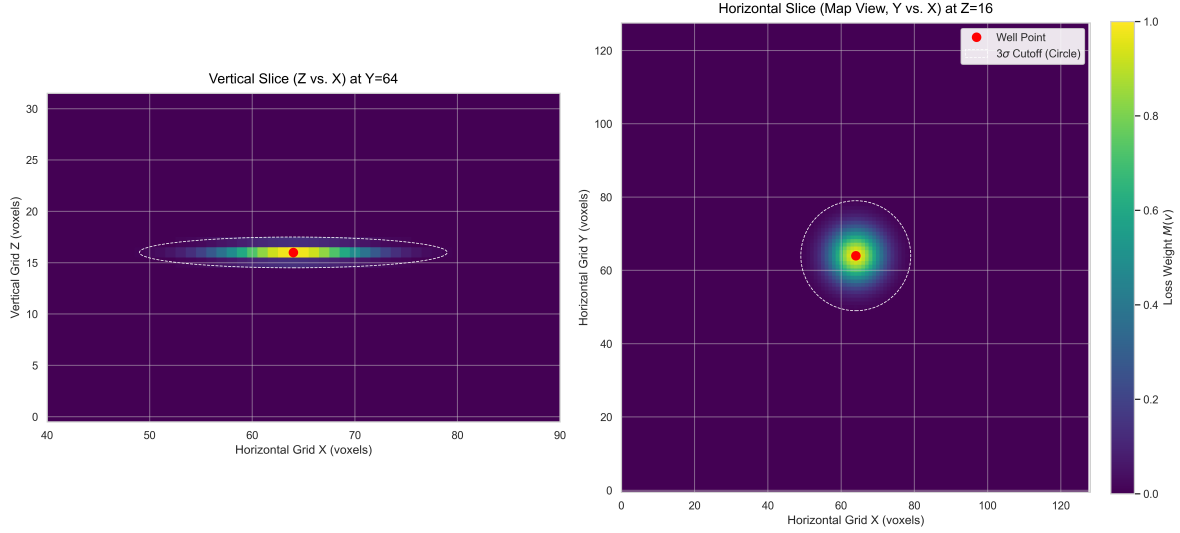


Figure C.1: Orthogonal cross-sections illustrating the 3D anisotropic Gaussian weight mask $M(v)$ applied during Latent Space Optimization. The anisotropy factors are 10,1,1. The distance threshold (σ) controls the smooth spatial decay of the penalty away from the well point (red dot). Due to the anisotropic scaling factor, the mask forms a horizontal lens in the vertical cross-section (left), respecting stratigraphic deposition, while maintaining a perfectly radial spread in the map view (right). The gradient is safely truncated at a 3σ limit (dashed line) to preserve computational efficiency.

C.1.2. Pivotal Tuning

To overcome the inherent boundaries of the learned latent space, Pivotal Tuning (Roich et al., 2022) is applied. Using the previously optimized $\mathbf{z}^*$ as a static anchor, the generator’s internal weights θ are unfrozen and optimized:

$$\theta^* = \arg \min_{\theta} \mathcal{L}_{CE}(G_{\theta}(\mathbf{z}^*)_{\text{well}}, \mathbf{y}_{\text{well}}) + \lambda \|\theta - \theta_0\|_2 \quad (\text{C.3})$$

This formulation pushes the localized outputs to perfectly match the hard conditioning data. The critical addition here is the L_2 regularization term $\lambda \|\theta - \theta_0\|_2$, where θ_0 represents the original pre-trained weights and λ is a penalty coefficient. This term forces the modified weights to stay as close as possible to their original state, preserving the global geological realism of the model and preventing catastrophic forgetting.

Computational Scaling: Unlike latent optimization, Pivotal Tuning is computationally intensive. The optimization scales with the total number of trainable parameters in the generator, $|\theta|$, which often numbers in the tens of millions for 3D generative networks, $\mathcal{O}(|\theta| \cdot N_w)$. It requires backpropagation through the entire network architecture for every iteration, scaling linearly with the size of the 3D grid output during the forward pass.

C.2. Generative Evaluation Metrics

The evaluation framework relies on advanced statistical divergence and optimal transport metrics to overcome the limitations of standard voxel-by-voxel assessments in unconditioned 3D modeling.

C.2.1. Wasserstein Distance and Mean Sliced Wasserstein Distance

The conceptual foundation for structural evaluation is the Wasserstein distance, which quantifies the optimal transport cost required to reconfigure the generated probability distribution P_g into the reference real data distribution P_r (Panaretos & Zemel, 2019; Rongier & Peeters, 2025b). It is defined as:

$$\mathcal{W}(P_r, P_g) = \inf_{\gamma \sim \Pi(P_r, P_g)} \mathbb{E}_{(x,y) \sim \gamma} [\|x - y\|] \quad (\text{C.4})$$

where $\Pi(P_r, P_g)$ denotes the set of all joint probability distributions $\gamma(x, y)$ whose marginals are P_r and P_g respectively.

Because computing Equation C.4 over high-dimensional 3D grids suffers from the curse of dimensionality, we employ the Mean Sliced Wasserstein Distance (MSWD). MSWD projects the complex 3D distributions onto 1D lines where the optimal transport problem has a trivial, closed-form solution (sorting). Mathematically, it is defined by integrating the 1D Wasserstein distances over the unit hypersphere $\mathbb{S}^{d-1}$:

$$\text{MSWD}(P_r, P_g) = \int_{\mathbb{S}^{d-1}} \mathcal{W}_{1D}(\mathcal{R}_\omega \# P_r, \mathcal{R}_\omega \# P_g) d\omega \quad (\text{C.5})$$

where $\mathcal{R}_\omega \# P$ denotes the projection (Radon transform) of distribution P onto a 1D line defined by the direction vector ω . In practice, this integral is approximated using a Monte Carlo approach by averaging over a large number of random projection angles L .

Computational Scaling: Computing the exact Wasserstein distance scales at roughly $\mathcal{O}(V^3 \log V)$, where V is the total number of voxels in the 3D grid. This is intractable for standard geomodels. MSWD circumvents this by scaling at $\mathcal{O}(L \cdot V \log V)$, where L is the number of projection slices. Since sorting 1D arrays is exceptionally fast, MSWD allows for robust structural comparison of massive 3D grids in a fraction of the time.

C.2.2. Uncertainty and Spatial Divergence Metrics

To assess spatial uncertainty and validate generation diversity across an ensemble, Normalized Shannon's Entropy (Shannon, 1948) is calculated independently for every spatial coordinate (voxel) $\mathbf{v} \in \mathbb{R}^3$. It scales the standard entropy equation by the number of geological classes (k) to bound the metric between 0 and 1:

$$H_{\text{norm}}(\mathbf{v}) = -\frac{1}{\log_2(k)} \sum_{i=1}^k P(F_i|\mathbf{v}) \log_2 P(F_i|\mathbf{v}) \quad (\text{C.6})$$

Here, $P(F_i|\mathbf{v})$ is the empirical probability of generating facies F_i at the specific voxel $\mathbf{v}$ across the entire ensemble of realizations. An $H_{\text{norm}}(\mathbf{v})$ of 0 implies total certainty (uniformity or potential mode collapse at that pixel), while 1 implies total uncertainty (maximum diversity across all k facies).

Computational Scaling: Entropy calculations are highly efficient parallel operations. The scaling is purely linear, $\mathcal{O}(V \cdot k)$, where V is the total number of voxels, making it perfectly suited for massive 3D grids regardless of ensemble size.

To quantify the distributional disagreement between the true reference models (P) and the generated models (Q), we calculate the pixel-wise Kullback-Leibler (KL) divergence (Kullback & Leibler, 1951):

$$D_{KL}(P_{\mathbf{v}} \parallel Q_{\mathbf{v}}) = \sum_{i=1}^k P(F_i|\mathbf{v}) \log_2 \left(\frac{P(F_i|\mathbf{v})}{Q(F_i|\mathbf{v})} \right) \quad (\text{C.7})$$

Because KL divergence is asymmetric and unbounded, it is transformed into the Jensen-Shannon (JS) divergence (Lin, 1991) to provide a bounded, symmetric measure of stochastic spread at each voxel:

$$D_{JS}(\mathbf{v}) = \frac{1}{2} D_{KL}(P_{\mathbf{v}} \parallel M_{\mathbf{v}}) + \frac{1}{2} D_{KL}(Q_{\mathbf{v}} \parallel M_{\mathbf{v}}) \quad (\text{C.8})$$

where $M_{\mathbf{v}} = \frac{1}{2}(P_{\mathbf{v}} + Q_{\mathbf{v}})$ is the mixture distribution at voxel $\mathbf{v}$.

Finally, to aggregate this 3D grid of spatial conflict into a single definitive scalar for model evaluation, the mean JS divergence is computed across the entire spatial volume $\mathcal{V}$:

$$\overline{D}_{JS} = \frac{1}{|\mathcal{V}|} \sum_{\mathbf{v} \in \mathcal{V}} D_{JS}(\mathbf{v}) \quad (\text{C.9})$$

Computational Scaling: Similar to entropy, JS Divergence operates strictly on the pre-computed probability mass functions of the respective ensembles. Therefore, the scalar aggregation scales linearly with the number of voxels evaluated, $\mathcal{O}(V)$. It remains computationally trivial even for massive, high-resolution spatial domains.

To aggregate this spatial conflict into a single quantitative scalar comparing two datasets, we utilize the Jensen-Shannon (JS) divergence (Lin, 1991). It measures the symmetric stochastic spread between the true reference distributions (P) and the generated distributions (Q):

$$D_{JS}(P \parallel Q) = \frac{1}{2}D_{KL}(P \parallel M) + \frac{1}{2}D_{KL}(Q \parallel M) \quad (\text{C.10})$$

where $M = \frac{1}{2}(P + Q)$ is a mixture distribution, and D_{KL} is the standard Kullback-Leibler divergence (Kullback & Leibler, 1951), explicitly formulated as:

$$D_{KL}(P \parallel Q) = \sum_x P(x) \log \left(\frac{P(x)}{Q(x)} \right) \quad (\text{C.11})$$

Unlike D_{KL} , which is asymmetric and can approach infinity if $Q(x)$ is zero, D_{JS} is strictly symmetric ($D_{JS}(P \parallel Q) = D_{JS}(Q \parallel P)$) and bounded between $[0, 1]$ when calculated with base-2 logarithms, making it a robust final scalar for spatial fidelity.

Computational Scaling: Similar to entropy, JS Divergence operates strictly on the pre-computed probability mass functions of the respective ensembles. Therefore, it scales linearly with the number of spatial bins/voxels evaluated, $\mathcal{O}(V)$. It remains computationally trivial even for massive, high-resolution spatial domains.